

The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties

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A thesis submitted in fulfilment of the requirements of the degree of Doctor of Philosophy in Physics, the University of Auckland, 2024. This thesis is for examination purposes only and is confidential to the examination process.

Abstract

A short description of the work.

Cover Image: The mass profile of a $10 M_{\odot}$ star from zero-age main sequence until the end of the asymptotic giant branch. Author's own work.

For my grandparents: Alan and Nanette, and Walter and Eileen.

Swing low, sweet chariot.

Popatrz dookoła, nikt tu nie wie po co żyć
Look around, no one here knows what to live for

I o czym śnić
And what to dream about

Za sobą zostaw górę trosk i wyrusz zanim zgubisz trop
*Leave the mountain of worries behind you and set out before you
lose the trail*

– *Dzień dobry* by Kwiat Jabłoni

Acknowledgements

Ho there, wanderer! Stay thy course a moment and indulge an old man.

ELMINSTER AUMAR

Waipapa Taumata Rau (The University of Auckland) sits within the rohe of several Māori iwi/hapū, including Ngāti Whātua, Ngāti Whātua o Ōrākei, Te Kawerau a Maki, Ngāti Tamaoho, Te Ākitai Waiohū, Ngāti Maru, Te Patukirikiri, Ngāti Pāoa, Ngāti Tamaterā, Ngāi Tai ki Tāmaki, Ngāti Te Ata, and Ngāti Whanaunga, and Waikato-Tainui. Ipipiri (the Bay of Islands) sits within the rohe of Ngāpuhi and Ngāti Rēhia. I acknowledge these iwi and hapū as the tāngata whenua of Tāmaki Makaurau, Ipipiri, and Aotearoa, I acknowledge that sovereignty was never ceded under Te Tiriti O Waitangi, and I pay my respects to kuia and kaumātua of these iwi and hapū, past, present, and future. Tūturu whakamaui, kia tina! Tihei mauri ora!

The author wishes to acknowledge the use of New Zealand eScience Infrastructure (NeSI) high performance computing facilities, consulting support and/or training services as part of this research. New Zealand’s national facilities are provided by NeSI and funded jointly by NeSI’s collaborator institutions and through the Ministry of Business, Innovation & Employment’s Research Infrastructure programme. URL <https://www.nesi.org.nz>.

This work made use of v2.2 of the Binary Population and Spectral Synthesis (BPASS) models as described in [Eldridge et al. \(2017\)](#) and [Stanway & Eldridge \(2018\)](#).

Once a king or queen in Narnia, always a king or queen.

ASLAN

I am the proud grandchild of immigrants, and by my count, the first person who identifies as Sāmoan to graduate with a PhD from the Department of Physics at the University of Auckland in (at least) 23 years. I’m also one of the few gender non-conforming students in my Department. To all my fellow BIPOC students, all my queer whānau, and all my other underrepresented students in the sciences, I see you, I recognise you, I feel you. Kia kaha, kia māia, kia manawanui.

Who am I?

I am a girl who loves my island

And the girl who loves the sea, it calls me

I am the daughter of the village chief

We are descended from voyagers

Who found their way across the world

They call me

I Am Moana FROM “MOANA”

It would be impossible for me to list my entire whakapapa. It is known as far back as nine generations in parts, tracing back to around the mid-1700s – including to the first Lord Mayor of the City of Sheffield, a

Prime Minister of Samoa, a former Minister of Finance and Chair of Samoa's National Bank, and some of the major clans in Samoa. At last count we know of about 400 relatives – my Great-Great-Aunt's father-in-law had thirteen children in the mid-1800s. But I want to especially mihi to my mother Sue, my late aunt Leslie, my late grandfather Alan, my grandmother Nanette and her late mother Mary "Sugalu" Scott, my aunt Alison and uncle Pete, my great-uncle Ben and great-aunt Lulu, my cousins Jeremy, Jaime, Charlotte, Candice, Ngareta, Wairaka, Marihi, and Tama. To the rest of my 'aiga: e lautele le vateatea, ae le o le avanoa iinei: Ou te alofa ia te outou uma.

Life before death. Strength before weakness. Journey before destination.

FIRST IDEAL OF THE KNIGHTS RADIANT

For Jaffe, Bailey, Riegel, Ray, Mercer, Johnson, Willingham, and O'Brien: Vox Machina, the Mighty Nein, Bells Hells, Gilmore, Allura, Delilah, Pumat, Xandis, and Ryn. For Iyengar, Carazo, Ray, Riegel, Willingham, Wilson, and Mulligan: the Ring of Brass, Vespine, Maya, and Kir. For Lewis: Peter, Edmund, Susan, Lucy, Aslan and Jadis. For Sanderson: Kaladin, Syl, Shallan, Jasnah, Vin, Elend, Kelsier, Rysn, and Hoid. For Miyazaki: Totoro, Mei, Satsuki, Chihiro, Haku, and No-Face. For those who imagine the strange and beautiful and complex and weird and mystical worlds of fantasy, whether betwixt the pages of a book, on the silver screen, or upon the roll of a d20, I pay my enormous regards to you.

*Can you imagine a piece of the universe
More fit for princes and kings?
I'll trade you ten of your cities
For Marion Bridge and the pleasure it brings*

Song for the Mira BY JOHN McDERMOTT

It occurs to me that there are precious few moments in one's life to acknowledge and thank the people who contributed to where one is now. I hold the belief that a person is defined by the sum of their actions and interactions, so I would like to acknowledge a few of the people that have made me the person I am today.

I'd like at this point to acknowledge friends: Connor, Frazer, Girish, Hunter, Pierce, Ryan, Scott, and Sean, with whom I shared some of my first physics and mathematics courses. Josh, Tessa, Hamish, Jasmine, Ryan: my first experience with D&D was with Eldrin. Andrew, George, Hamish, Connor, Callum, and Mark: Katerina was my longest player experience; playing her taught me so much about myself (and Andrew). Josh, Liam, Arthur, Connor, Andrew, Oliver: Ayo and the rest of the Blue Originators (you never learned their party's name) owe you a debt of gratitude. Even good old Necromancer Bob says hi.

I've lost touch with some of the people I did high school with but the formative experiences there still sit inscribed in indelible ink in my mind. Caitlin, who has been one of my best friends for the better part of a quarter-century (and who would *not* let me live it down were I to forget her). Callum, Susannah, Kyle, Grayson, Tomek "if you're going to try to pronounce my last name, put some **EMPHASIS** on it" Pietkiewicz and the entire business and commerce crew. Sam, Louise, Crooks, the IT Ninjas. Isabella, Chyna, Evan, Parth, Jay, Matt, all the people at NZSDC 2015, even if I had to argue in favour of alternative medicine and exclusive schools for LGBTQIA youth, those moments will be with me forever. Finn and Susannah: I know they said they never ranked us in maths classes, but they ranked us in maths classes, and I know we always

vied for the top three places. Finn is an architect now and Susannah is admitted to the High Court – does that mean I finally win the ranking?

I'd also like to thank Di Eastwood, Maddy Haydock, Melissa Johansson, and Lauren Mulcahy (our debating coach, and my Year 13, 12, and 11 English teachers). Not all of you are teaching anymore, but I still remember vivid parts of your lessons. Miss Mulcahy commending my essay structure (for not opening an essay with "The question I am answering is..." but actually motivating why I enjoyed *To Kill a Mockingbird* and *Romeo+Juliet*). Miss Johansson, for showing faith in the process of my creative writing portfolio ("Sean, I want to send you to counselling for this story. Change nothing, this is an E."). Mrs Haydock, firstly for Mufasa but also for believing in my weird and experimentalist poetry (I was going through a bit of a fourth-wall breaking phase with poetry), for letting me rant for 30 minutes on why *The Matrix* is actually a Marxist allegory and the (concerningly frequent) themes from Freudian psychoanalysis we can pull from it. Moreover, I remember in English class, you congratulating me for using "moreover" correctly in a sentence. Mrs Eastwood for always challenging me to do better in debating. I've said it before and I will say it again: I credit a mix of you four for my erudition and eloquence.

On the topic of teachers, I would like to mihi to Robbie Lumkong, Andy Dyson, Chris Whitaker. Physics lessons in Year 12 and 13 with Mr Lumkong (and in Year 13 with Mr Dyson) were absolutely batshit crazy – can I say batshit in a thesis? – but it is the undeniable infective enthusiasm for physics that someone who has played international rugby¹ brings that made me even think about doing PHYSICS 120 when I came to university and had a free paper in first year. Mr Whitaker, I still remember sitting on a bus behind you in Year 11 Physical Science and asking you a tonne of questions about the big bang and the standard model.

Right, I'm almost done here, I swear.

Physics is a collaborative endeavor, and I would be remiss to not acknowledge my research compatriots. Thank you to the BPASS team, both in Auckland and abroad: John, Heloise, Max, Sohan, Wouter, Petra, Nicolas, Adib, Conor, Gareth, Ashley, and Ankur. We've also been joined by a few students who have since left the team (because they graduated!): Gleb, Emma, Louis, Ethan, Siris, Celina, and Amanda. Thank you to our PIs, Dr Elizabeth Stanway and Prof. Jan Eldridge.

Speaking of Jan! Thank you to Prof. Jan Eldridge and Assoc. Prof. Nick Rattenbury. I genuinely cannot image a more perfect supervision team. Both carry a large amount of mana, and I have learned an immense amount from each of them. You'll notice that all of my contractions of people's titles either omit (in the case of "Dr") or contain periods (in the case of basically every other title). Nick was very pedantic about this in my Masters thesis and now I subconsciously check for this when reading other people's prose. Thank you by extension to Prof. Richard Easther. Jan, Nick, and Richard were my lecturers for PHYSICS 107/G in Semester Two of 2016. I needed extra papers to fill out my first year, as there were only three required papers for my (at that time) Computer Science degree. I took PHYSICS 120 and 150 because I was good at it in High School (see Mr Lumkong, Mr Dyson, and Mr Whitaker above), and PHYSICS 107/G because it sounded fun. I left that course a physics double major, and no joke, restructured my entire degree around taking PHYSICS 356. Jan, Nick, and Richard are a large part of why I am a scientist today. I cannot thank them enough for that. From the bottom of my heart, thank you, ngā mihi, vinaka vakalevu, fa'afetai lava.

I'd also like to mihi to a few of my inspirational heroes in science as a whole. Prof. Stephen Hawking, Prof. Kip Thorne, Prof. Brian Greene, Prof. Brian Cox, Dr Derek Miller (Veritasium), Mrs Dianna Cower

¹This is not a joke or a quip, my Year 13 physics teacher played as a number eight in the 1999 Rugby World Cup for the USA.

(PhysicsGirl), Mr Destin Sandlin (SmarterEveryDay), Mr John D. Boswell (melodysheep).

*You'll be in my heart.
No matter what they say.
You'll be here in my heart.
Always.*

You'll be in my heart BY PHIL COLLINS

When I started this PhD, I had two alive grandparents. I now have only one, my grandmother Nanette. For Alan, Walter, and Eileen, I carry your memories with me always. In the words of Liam O'Brien: *I was richer for having known you, friend.*

I could not be more keen, for some plain old boring precedented times.

CR. RICHARD HILLS

I've noticed that the more theses I write, the more people I thank. My PGDipSci dissertation thanked five people. My Masters thesis thanked some fifty-five, a Government, and several coffee companies. Did I beat it this time?

Run, you clever [girl], and be a Doctor.

CLARA OSWIN OSWALD

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Glossary

Notation	Description
pp -chain	proton-proton chain.
$pp1$ -chain	path 1 proton-proton chain.
$pp2$ -chain	path 2 proton-proton chain.
$pp3$ -chain	path 3 proton-proton chain.
3D MHD	3D Magnetohydrodynamics.
BH	Black Hole.
BNS	Binary Neutron Star.
BPASS	Binary Population and Spectral Synthesis.
CCSN	core-collapse supernovae.
CDF	Cumulative Distribution Function.
CEE	Common Envelope Evolution.
CME	coronal mass ejection.
DTD	Delay-Time Distribution.
GR	General Relativity.
IMF	Initial Mass Function.
ISM	Interstellar Medium.
JWST	James Webb Space Telescope.

Notation	Description
K-S	Kolmogorov-Smirnov.
LIGO	Laser Interferometer Gravitational Wave Observatory.
LISA	Laser Interferometer Space Antenna.
LVK Collaboration	LIGO-Virgo-KAGRA Collaboration.
MLT	Mixing Length Theory.
MWEG	Milky-Way Equivalent Galaxy.
NS	Neutron Star.
NS-BH	Neutron Star - Black Hole Binary.
O3	Operating Run 3.
O4	Operating Run 4.
PDF	Probability Density Function.
SFH	Stellar Formation History.
SFRD	Star Formation Rate Density.
SPH	Smoothed-particle hydrodynamics.
USSN	ultra-stripped supernova.
WD	White Dwarf.
ZAMS	Zero-Age Main Sequence.

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2.3 if this bit stays, 92% of the sun being hydrogen and 0.2% being iron needs a source	10
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Thesis Epigraphs

If the stars should appear one night in a thousand years, how would men believe and adore; and preserve for many generations the remembrance of the city of God which had been shown! But every night come out these envoys of beauty, and light the universe with their admonishing smile. (<i>Nature</i> , by Ralph Waldo Emerson)	2
Come with me, and you'll be, in a world of pure imagination. (<i>Pure Imagination</i> , by Gene Wilder)	5
Se'i lua'i lou le 'ulu taumamao. (Pick the breadfruits on the far-off branches first) (Sāmoan proverb)	7
Nature and nature's laws lay hid in night: God said, <i>Let Newton be!</i> and all was light. (Alexander Pope)	7
It did not last: the Devil howling "Ho! Let Einstein be!" restored the status quo. (<i>Answer to Pope's Epitaph for Sir Isaac Newton</i> , by J. C. Squire)	8
Pallida mors æquo pulsat pede pauperum tabernas Regumque tures.	
Pale death knocks with the same tempo upon the huts of the poor and the towers of kings. (Horace)	13
Algebra's like sheet music. The important thing isn't can you read music, it's can you <i>hear</i> it. Can you hear the music, Robert? (<i>Niels Bohr</i> , in Oppenheimer)	16
Wit nodded. "Oh, is this the place where I talk?" Kaladin said. "If you wish." "I don't wish. Get on with the story." (<i>Rhythm of War</i> , by Brandon Sanderson)	22
I think there is a world market for maybe five computers. (Thomas Watson, President of IBM, 1943)	24

And through the night, behind the wheel, the mileage clicking west
 I think upon Mackenzie, David Thompson and the rest
 Who cracked the mountain ramparts and did show a path for me
 To race the roaring Fraser to the sea.
 (*Northwest Passage*, by Stan Rogers) 24

Looking up, he saw a thousand stars glittering in the deep velvet of a night with no moon. He knew
 them all, their stories and their names. He knew them in a familiar way, the way he knew his own
 hands.
 (*The Name of the Wind*, by Patrick Rothfuss) 26

Begin at the beginning, the King said, very gravely, and go on till you come to the end: then stop.
 (*Alice’s Adventures in Wonderland*, by Lewis Carroll) 27

Captain Holt: It works! I’ve never been more proud of you for anything in your life.
Terry: I mean, I’ve solved a lot of cases for you.
Captain Holt: And yet crime has continued.
 (Brooklyn Nine-Nine) 27

All hands, this is the Captain. Begin prepping all stations for first contact.
 (Captain Ed Mercer, “The Orville”) 27

I fear not the man who has practiced 10,000 kicks once,
 but I fear the man who has practiced one kick 10,000 times.
 (Bruce Lee) 31

If you torture the data enough, they will confess.
 (Salvatore Vitale) 31

Oh! I have slipped the surly bonds of Earth
 And danced the skies on laughter-silvered wings;
 ...
 And, while with silent lifting mind I’ve trod
 The high untrespassed sanctity of space,
 Put out my hand, and touched the face of God.
 (*High Flight*, by John Magee Jr.) 47

Take-off is optional. Landing is mandatory.
 (First rule of aviation) 47

Though in reviewing the incidents of my administration I am unconscious of intentional error, I am nevertheless too sensible of my defects not to think it probable that I may have committed many errors.

(George Washington) 47

Co-Authorship Form

This form is to accompany the submission of any PhD that contains published or unpublished co-authored work. **Please include one copy of this form for each co-authored work.** Completed forms should be included in all copies of your thesis submitted for examination and library deposit (including digital deposit), following your thesis Acknowledgements. Co-authored works may be included in a thesis if the candidate has written all or the majority of the text and had their contribution confirmed by all co-authors as not less than 65%.

Please indicate the chapter/section/pages of this thesis that are extracted from a co-authored work and give the title and publication details or details of submission of the co-authored work.	
Chapter 5 is based on work from “New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations”, published in the Monthly Notices of the Royal Astronomical Society (MNRAS), here: dx.doi.org/10.1093/mnras/stad977	
Nature of contribution by PhD candidate	Wrote majority of the paper, developed most parts of the testing pipeline, later data generation, performed the statistical tests.
Extent of contribution by PhD candidate (%)	90

CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Original data generation, guidance on project, wrote clarification on the difference between BPASS primary and secondary models, criticism on manuscript
Max Briel	Developing ideas, code commentary, criticism on manuscript
Heloise Stevance	Developing the test for Ultra-Stripped Supernovae, guidance on project, criticism on manuscript
Reinhold Willcox	Providing single-star pulsar dataset from Willcox+ 2021, criticism on manuscript

Certification by Co-Authors

The undersigned hereby certify that:

- ❖ the above statement correctly reflects the nature and extent of the PhD candidate’s contribution to this work, and the nature of the contribution of each of the co-authors; and
- ❖ that the candidate wrote all or the majority of the text.

Name	Signature	Date
Jan Eldridge		
Max Briel		
Heloise Stevance		
Reinhold Willcox		

Introduction

*Una volta che avrai
Spiccato il volo, deciderai
Sguardo verso il ciel saprai:
Lì a casa il cuore sentirai.*

LEONARDO DA VINCI

*Once you have taken flight
You'll decide
Gaze towards the sky, you'll know that:
That is where your heart will feel at home.*

LEONARDO DA VINCI

On September 14, 2015, the Laser Interferometer Gravitational Wave Observatory (LIGO) Collaboration made the first direct detection of gravitational waves, from the merger of a binary black hole with masses $m_1 = 36^{+5}_{-4} M_\odot$ and $m_2 = 29^{+4}_{-4} M_\odot$ (Abbott et al., 2016). This detection sparked the so-called “gravitational wave era”, with detections in Operating Run 4 (O4) reading a cadence of hundreds per year. These detections afford researchers probes of various physical phenomena. For instance, these merger events are some of the most energetic events in the Universe (GW150914 released over 50 times the energy of all stars in the observable Universe, per Abbott et al. (2016)), and the merger of binary neutron stars are the predominant formation channels for most of the elements heavier than iron (see, among others, Eichler et al., 1989; Arcavi et al., 2017; Pian et al., 2017; Smartt et al., 2017). In addition, the stochastic background of gravitational waves is a direct probe on early-Universe cosmology (Christensen, 2018), though the frequency band that these gravitational waves span is between 10^{-9} and 10^{14} Hz (Christensen, 1992, 2018), making detection of the stochastic background difficult (Agazie et al., 2023; Antoniadis et al., 2023; Reardon et al., 2023; Xu et al., 2023).

Any attempt at understanding gravitational waves must be founded in a strong theoretical understanding of stellar structure and evolution. Without theory, any observation is no more than a curiosity; it is with theoretical underpinning and models that insight can be gained. As most astrophysical sources of gravitational waves (i.e., not the stochastic background) are formed by the merger of two “compact remnants” – black holes, white dwarfs, and neutron stars – one must understand their formation, and thus the evolution of progenitor stars into these remnants, as well.

Furthermore, a sound understanding of the evolution of stars has impacts beyond gravitational wave astronomy. As the interiors of stars are complex nuclear furnaces, they are responsible for producing most of the elements in the Universe heavier than Hydrogen (Burbidge et al., 1957). In fact, of the elements lighter than Uranium, the only elements not produced in stellar processes (such as through kilonovae or in supernovae) are technetium, promethium (which both lack stable isotopes), boron, beryllium (which both are produced via cosmic ray fission), and some formation channels of lithium. Stars are also the focal point of planetary systems, and provide the theoretical right conditions for the formation of life.

However, this diversity and complexity of stellar interiors demands a careful and varied approach. One cannot abstract away much of the important physics. Although it is common to use approximations such as spherical symmetry to reduce the dimensionality of the problem, and to only evaluate the equations of stellar structure at a fixed number of locations in this one remaining dimension, chosen sufficiently intelligently, a complex computational effort must be undertaken.

*If the stars should appear one night in a thousand years, how would
men believe and adore; and preserve for many generations the
remembrance of the city of God which had been shown! But every
night come out these envoys of beauty, and light the universe with
their admonishing smile.*

Nature, BY RALPH WALDO EMERSON

Observational astronomy can essentially be traced back to the earliest cultures. Babylonia knew of Halley's comet and tracked observations of it as late as 87 BCE ([Stephenson et al., 1985](#)). Records of eclipses in Chinese culture exist as far back as 1600 BCE ([Fang, 2015](#)). Knowledge of the movements of stars guided Polynesian wayfarers as they crossed the South Pacific ([Buck, 1938](#); [Hamacher, 2022](#)).

Even in the arts, the knowledge of the lunar perigee was written into *Othello* by Shakespeare:

It is the very error of the moon;
She comes more near the earth than she was wont,
And makes men mad.

– [Shakespeare \(ca. 1603\)](#)

Knowledge of oceanic tides was posited by Ptolemy, writing that “the tides of the sea are ruled by [the moon's] risings and settings” (original work [Ptolemy & Proclus \(ca. 168\)](#), summary provided by [Tabarroni \(1989\)](#)), which influenced the writings of Dante:

E come 'l volger del ciel de la luna
cuopre e discuopre i liti senza posa,
così fa di Fiorenza la Fortuna:

And, as the turning of the lunar sphere covers
and endlessly uncovers the edges of the shore,
thus does fortune deal with Florence.

– [Alighieri \(ca. 1317\)](#)

In religious scripture, celestial navigation has oft been a fixture, such as in Chapter 2 of the Gospel according to Matthew:

2:7 – Then Herod, when he had priuily called the Wise men, enquired of them diligently
what time the Starre appeared:

2:8 – And he sent them to Bethlehem, and said, Goe, and search diligently for the yong

child, and when ye haue found him, bring me word againe, that I may come and worship him also.

2:9 – When they had heard the King, they departed, and loe, the Starre which they saw in the East, went before them, till it came and stood ouer where the young childe was.

– Various Authors (1611)

Despite this deep cultural connection to the stars, what we would now call Astronomy is a somewhat recent invention. Arguably it arose in the 16th century CE with Copernicus proposing the heliocentric model of the solar system, which was confirmed by Galileo in the 17th century CE (Carroll & Ostlie, 2017) – although Galileo’s claims did result in his being convicted of heresy for 300 years.

The early 20th century saw a veritable renaissance in both the observational and theoretical spheres. On the one hand, developments such as the theories of Special Relativity in 1905 (Einstein, 1905), General Relativity (GR) in 1917 (Einstein, 1917), and the predictions of gravitational waves (Poincaré, 1906; Einstein, 1916) led to theoretical revolutions. On the other hand, the innovations of Payne (1925) in showing that stars are composed of mostly Hydrogen and Helium, Baade & Zwicky (1934) in distinguishing “super-novae” from “common novae”, Hoyle (1946) in developing the foundation of the nuclear reaction of elements heavier than Hydrogen in stars, Salpeter (1955) in developing the Initial Mass Function (IMF), Burbidge et al. (1957) in truly developing the theory of stellar nucleosynthesis, Henyey et al. (1955) and Hayashi (1961) in exploring the pre-main sequence evolution of stars were key to kickstarting entire branches of astronomy such as spectroscopy and computational astrophysics.

The pace of discovery of new and exotic astronomical phenomena has not slowed into the 21st century. Thorne-Żytkow objects – giant stars with Neutron star cores (Thorne & Zytkow, 1977) – once only existed as theoretical curiosities, yet now have credible observational candidates in Levesque et al. (2014). The aforementioned gravitational waves were detected in 2015 (LIGO Scientific Collaboration and Virgo Collaboration, 2016), a century after they were demonstrated by Einstein (1916) to be a central prediction of GR. More recently, Lau et al. (2022) directly imaged 17 carbon-rich dust rings around WR140 with James Webb Space Telescope (JWST), enriching the Interstellar Medium (ISM) with carbon-rich compounds. In the coming years, the Vera C. Rubin Observatory (Ivezić et al., 2019) and Laser Interferometer Space Antenna (LISA) are expected to vastly increase the number of short-period eclipsing double white dwarfs (Burdge et al., 2020), with the latter able to detect gravitational waves with frequencies between 0.1 mHz and 1 Hz (Amaro-Seoane et al., 2012), compared to the ~ 10 Hz – 1000 Hz that LIGO-Virgo-KAGRA Collaboration (LVK Collaboration) can detect (Schmitz, 2021).

Astronomy has evolved from the era of the naked-eye, to the era of the telescope, to the era of the observatory, and now the gravitational wave era. The next era of astronomy is the big data era. With observations reaching sub-day cadences, a large array of theoretical models, and perhaps the development of “on-demand” detailed models is required.

1.2

Stellar Modelling

Modelling the evolution of a star is a nontrivial endeavour. Stars as objects obey GR owing to their masses and sizes, and reach internal temperatures, pressures, and densities unsustainable on Earth. For instance,

the core of a star is capable of nuclear fusion, which has not been recreated on Earth. Moreover, the density of Sirius B (a White Dwarf (WD) in the constellation of Canis Major) is $2.14 \times 10^{10} \text{ kg m}^{-3}$ (Barstow et al., 2005) and the mean density of the Sun is $\sim 1410 \text{ kg m}^{-3}$ (Eddy & Ise, 1979). On Earth, the highest density measured in a laboratory is $\sim 27\,000 \text{ kg m}^{-3}$, requiring pressures on the order of 2.3 TPa (Fratanduono et al., 2020), and observed for only nanoseconds.

Because of these factors, one often turns to computer simulations to model the interiors of stars, as it is nigh impossible to achieve and sustain these conditions to experimentally test any physical phenomena. That said, missions such as the Parkes Solar Probe [TODO: cite] have made flyby missions into the atmosphere of the Sun, though broadening the sample size to two stars – using Proxima Centauri, the nearest star to Earth aside from the Sun – would require travelling at approximately $0.1c$ for 40 years to reach. If instead we were to send the Parker Solar Probe travelling at a constant velocity of its top velocity of $\sim 176\,471 \text{ m s}^{-1}$, it would take approximately 7200 years to reach Proxima Centauri. Needless to say, physical experimentation with a star is difficult.

1.2.1

Population Synthesis Codes

Broadly speaking, there are three approaches to modelling stars, all of which have both merits and drawbacks. These three methods are:

1. Rapid Population Synthesis,
2. Detailed Population Synthesis, and
3. 3D Magnetohydrodynamics (3D MHD) Simulations.

Each of these approaches informs the other two, and all play important roles in stellar modelling. Although we focus on detailed methods, we will briefly outline all three approaches here.

Detailed Population Synthesis

Detailed stellar models explicitly solve the equations of hydrostatic equilibrium, usually via some form of matrix-inversion method. For instance, as will be discussed in subsection 2.7.1, the Cambridge STARS code inverts the Henyey matrix $\mathbf{H}_{ij} = \frac{\partial(\text{equation } i)}{\partial(\text{variable } j)}$. MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023) uses a Newton-Raphson solver in order to converge to a solution. In particular, we use a variant of the Cambridge STARS code, which solves the equations of *both* stars simultaneously.

Rapid Population Synthesis

Rapid population synthesis methods exploit the statistical properties of stars in order to rapidly infer properties of populations of stars. However, they still require as input detailed population synthesis models – for example, COMPAS (Team COMPAS: Riley et al., 2022) uses MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023) models as its input models.

3D MHD Simulations

3D Magnetohydrodynamics (3D MHD) and Smoothed-particle hydrodynamics (SPH) simulations are the most detailed method of stellar modelling. The base principle is that the star is evaluated as a network of particles in a continuous media, and the evolution of the star is followed in theoretically atomic detail. For example, PHANTOM (Price et al., 2018) is a SPH code which essentially models the star as a fluid, where the Lagrangian hydrodynamical equations are solved for in two-dimensional cells (termed “particles”). Although these present the most detailed models, they are also the most time-intensive to solve. For a SPH code, the famous Navier-Stokes equation is invoked, which has no analytical solution and is computed in excruciating detail. As a result, most SPH models take weeks to run and simulate a star for kiloseconds of its lifetime, rather than for the millions of years detailed models are able to compute.

Though it may seem that they serve the same niche and are therefore competing to be the “better” method of stellar modelling, detailed 1D models and 3D MHD/SPH models exist harmoniously with each other. The advancements in 3D MHD/SPH modelling inform the evolution of 1D models in a process known as “3-to-1”, and the computational flexibility of 1D models offers insights into where 3D models wish to target to evolve.

1.3

Thesis Outline

Come with me, and you'll be, in a world of pure imagination.

Pure Imagination, BY GENE WILDER

This thesis is, broadly, a tour of stellar evolution from Zero-Age Main Sequence (ZAMS) to the eventual merger of two compact objects, with a few omissions along the way (such as the moment of supernova). The thesis is structured as follows. In Chapter 2, we provide the necessary theoretical background for future chapters. In Chapter 3, we critically examine the pieces of software and numerical methods employed in this work, and discuss the logistics of running this single-thread grid-based code as a multiprocessed code on a distributed cluster. In Chapter 4, we present the results of our stellar evolution codes: both as results for single and binary star evolution, as well as the interesting case when both stars almost (or in some instances, do) touch. In Chapter 5, we examine the kinematics of the post-supernova inspiral and the mechanics of the natal supernova kick. In Chapter 6, we review the thesis and conclude. Finally in the appendices, we provide the KAITIAKI code necessary to generate the models for the stellar evolution plots throughout the thesis.

Theoretical Background

Se'i lua'i lou le 'ulu taumamao. (Pick the breadfruits on the far-off branches first)

SĀMOAN PROVERB

In this chapter, we provide a brief account of the evolution of a star from a theoretical perspective, from its formation to post-supernova. We will discuss both the case of a single isolated star, and an isolated binary. It is pertinent to note that our binaries will be assumed to have formed at birth, rather than by, for example, the dynamical capture of a runaway star in a globular cluster, which introduces extra effects and considerations. This process can be modelled by evolving a single star to a certain point, and then enabling binary evolution.

2.1 Stellar Formation

Nature and nature's laws lay hid in night: God said, Let Newton be! and all was light.

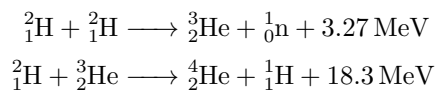
ALEXANDER POPE

From the point of view of stellar evolution, which starts in earnest at the ZAMS, the formation of stars is interesting but ultimately broadly unimportant. We provide a short recap of stellar formation here as scene-setting, but our stellar evolution models assume that a gas cloud has already collapsed and we have a star at its ZAMS.

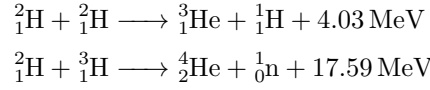
Stars are born in clouds of dust in rich H(II) regions of space. Due to inhomogeneities in the dust formation, some regions become more dense than others. When a region of this gas with adiabatic constant γ and mean molecular weight μ reaches the so-called Jeans Mass,

$$M_J = \left(\frac{10k_B T}{3(\gamma - 1)\mu m_H G} \right)^{\frac{3}{2}} \left(\frac{3}{4\pi\rho} \right)^{\frac{1}{2}} \sim \left(\frac{10k_B T}{G} R \right)^{\frac{3}{2}} \left(\frac{1}{M} \right)^{\frac{1}{2}} \quad (2.1)$$

where the similarity is taking $\gamma = \frac{5}{3}$, $\mu = \frac{1}{2}$ for an adiabatic gas composed of H(II), then gravity becomes the dominant force in the clumps of gas. This causes a protostar to form. This protostar slowly gains more mass from its surroundings and, as it does so, increases in temperature. When the core temperature reaches 10^7 K, the hydrogen in the core begins to fuse into helium via the reaction chain



This is aided by two further hydrogen to helium reactions:



or, via a master equation,



with intermediate products which remain stable and are thus omitted for simplicity. This energy release will sustain the star throughout its lifetime, providing outwards pressure resisting gravity pulling the star inwards, in a balance termed *hydrostatic equilibrium*. Once hydrogen fusion has begun, we say that the star has reached ZAMS, and is now on the *main sequence*, where it will spend approximately 90% of its lifetime.

2.2 Stellar Evolution

It did not last: the Devil howling "Ho! Let Einstein be!" restored the status quo.

Answer to Pope's Epitaph for Sir Isaac Newton, BY J. C. SQUIRE

Now that our protostar has evolved onto the main sequence, it has become a star in its own right. Over the next few million years, it will continue to burn hydrogen into helium, forming a Helium core. It is at this point that the evolution encounters a number of bifurcation points, depending on the mass and chemical composition of the star. In general, while the star lives, the four equations of stellar structure hold:

$$\left. \begin{aligned}
\frac{dm}{dr} &= 4\pi r^2 \rho \\
\frac{dp}{dr} &= -\frac{GM\rho}{r^2} \\
\frac{dT}{dr} &= -\frac{3\kappa\rho L}{64\pi r^2 \sigma T^3} \\
\frac{dL}{dr} &= 4\pi r^2 \rho \epsilon
\end{aligned} \right\} \quad (2.3)$$

where ϵ is the rate of energy eneration per unit mass, κ is the opacity of the material, σ is the Stefann-Boltzmann constant, and all other symbols have their usual meanings.

These equations represent conservation of mass, hydrostatic equilibrium, energy transport, and thermal equilibrium, respectively. Knowing constants ϵ, κ , we have four equations for four unknowns. It is usually convenient to take the *Rosseland mean* opacity,

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu \quad (2.4)$$

where B is the black-body intensity at frequency ν and temperature T . The function κ_ν is obtainable from nuclear physics: it is intimately related to the scattering cross-section σ :

$$\frac{1}{\kappa_\nu \rho} = \frac{1}{\sigma n} \quad (2.5)$$

where ρ is the mass density of the star, and n is the particle number density.

On the other hand, ϵ is obtainable from knowing the nuclear reactions occurring in the star's core:

$$\epsilon = \sum_i^N \text{energy output of reaction } i \quad (2.6)$$

All told, in the simplest sense, the internal *structure* of the star can be modelled by hand by solving Equation 2.3 from $m = 0$ to $m = M$.

2.2.1 The chemical composition of stars

A star contains at its core a complex nuclear furnace. Since stars are born in regions of space rich in H(II) – molecular hydrogen – and obtain high enough pressures in the core to ignite fusion, the star will be sustained by the burning of predominantly Hydrogen and Helium. We thus group the three main stages into Hydrogen, Helium, and Metals (which we define as anything with atomic number ≥ 3 , i.e., Lithium and heavier elements). It is convenient to speak of the mass fraction of the element, i.e., the fraction by mass of a star that is a certain element. For our purposes, this is expressible as

$$X + Y + \sum_{i=0}^{115} \mathcal{Z}_i = 1, \quad (2.7)$$

where X is the mass fraction of Hydrogen, Y of Helium, and $\mathcal{Z}_i$ as the metal¹ with atomic number $i + 3$. When speaking about the chemical composition of a star it is also convenient to speak of the *metallicity*, the fraction at ZAMS of the star that is metal. For this, we define the metallicity as

$$Z \equiv \sum_{i=0}^{115} \mathcal{Z}_i, \quad (2.8)$$

such that Equation 2.7 becomes

$$X + Y + Z = 1. \quad (2.9)$$

Hydrogen burning: the *pp*-chain

A star spends $\sim 90\%$ of its lifetime on the main sequence, which is where it burns protons (bare hydrogen atoms) into helium. This reaction is called the proton-proton chain (*pp*-chain), and can be broken up into three potential paths. All three paths begin with the reaction **[TODO: finish]**

1. path 1 proton-proton chain (*pp1*-chain):
2. path 2 proton-proton chain (*pp2*-chain):
3. path 3 proton-proton chain (*pp3*-chain):

¹It is custom in astronomy to refer to anything with an atomic number greater than 2 as a metal, rather than the conventional definition of a material conductive at absolute zero, including most of the d-block, s-block, f-block, and about half of the p-block on the periodic table.

The metallicity of the Sun

It is often convenient to index our metallicities relative to that of the Sun. To do this, we of course need to know the chemical composition of the Sun. We ignore virtually every isotope of every element in favour of $^{56}_{26}\text{Fe}$ in particular. This is for a few reasons: it is the most stable element of fusion (nuclides higher than $^{56}_{26}\text{Fe}$ will undergo fission to it and nuclides lower than $^{56}_{26}\text{Fe}$ will fuse towards it), it is very abundant observationally, and its absorption lines in spectra are easily distinguishable. Thus we can base our metallicity calculations via a quantity known as the iron-abundance ratio,

$$[\text{Fe}/\text{H}] = \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right) \Big|_{\text{sun}}, \quad (2.10)$$

where N_{H} (N_{Fe}) is the number of Hydrogen (Iron) atoms per unit volume of the sun or star. Mechanically this is represented by

$$N_{\text{X}} = N_{\text{A}} \frac{M}{m_{\text{X}}} \quad (2.11)$$

where N_{A} is Avogadro's constant ($\sim 6.022 \times 10^{23} \text{ mol}^{-1}$), M is the mass of element X in the star, and m_{X} is the relevant molar mass. For instance, imagining a hypothetical iron-rich star of mass M consisting of 50% hydrogen and 50% iron, the $[\text{Fe}/\text{H}]$ would be

$$\begin{aligned} [\text{Fe}/\text{H}] &= \log_{10} \left(\frac{N_{\text{A}} \frac{M_{\text{Fe}}}{m_{\text{Fe}}}}{N_{\text{A}} \frac{M_{\text{H}}}{m_{\text{H}}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{N_{\text{A}} \frac{M_{\text{Fe}}}{m_{\text{Fe}}}}{N_{\text{A}} \frac{M_{\text{H}}}{m_{\text{H}}}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{M_{\text{Fe}} m_{\text{H}}}{m_{\text{Fe}} M_{\text{H}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{M_{\text{Fe}} m_{\text{H}}}{m_{\text{Fe}} M_{\text{H}}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{0.5 M m_{\text{H}}}{m_{\text{Fe}} \times 0.5 M} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{0.002 M_{\odot} m_{\text{H}}}{m_{\text{Fe}} \times 0.92 M_{\odot}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{m_{\text{H}}}{m_{\text{Fe}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{0.002 M_{\odot} m_{\text{H}}}{m_{\text{Fe}} \times 0.92 M_{\odot}} \right) \Big|_{\text{sun}} \\ &\sim 2.66 \end{aligned}$$

[TODO: if this bit stays, 92% of the sun being hydrogen and 0.2% being iron needs a source]

In this regime, we can express the metallicity as approximately

$$\mathcal{Z} = 10^{[\text{Fe}/\text{H}]} \mathcal{Z}_{\odot}, \quad (2.12)$$

and thus by knowing the simple iron content of the Sun or star, we can approximate the metallicity. Our iron-rich “star” would thus have a metallicity of approximately 460 times that of the sun.

Asplund et al. (2021) presents an estimate **[TODO: finish sentence]**

2.2.2

Evolutionary Pathways

In this section, we examine the four major regimes of stellar evolution for single-stars. The precise boundaries are debatable **[TODO: cite?]**, but are generally accepted to be logarithmically spaced.

Low-Mass Stars: $0.08 M_{\odot} < M < 2 M_{\odot}$

Low-mass stars ignite helium degenerately in their cores, leading to thermal runaway. The star can peak at a luminosity close to $10^{11} L_{\odot}$ (approximately 100 times the peak luminosity of SN 1987A, per Figure 5 of [Arnett et al. \(1989\)](#)). However, this peak in luminosity is attributable to nuclear fusion in the core, and the energy does not reach the surface, let alone escape into the ISM – rather, it is absorbed into the outer layers of the star. During this extremely brief thermal runaway, the luminosity begins to settle and the star is able to continue evolving. Models of low mass stars fail at this stage and care is required to evolve them.

After the ignition of helium, these stars cross the so-called Hertzsprung Gap – the region where very few stars are observed empirically. This region is bounded (on a colour-magnitude diagram, which is a transform of the HR diagram)² by the points $(M_v, B - V) = \{(2, 0.5), (-7, 0.1), (-6, 1.6)\}$ ([Hoyle, 1960](#)), where M_v is the V-band magnitude (magnitude in the visual band), and $B - V$ is the colour index of the star. The phenomenon of a gap existing is attributable to the fact that stars cross this gap exceedingly quickly: whereas it spends $\sim 90\%$ of its life on the main sequence, it spends $\sim 2\%$ of its life crossing the Hertzsprung gap ([Hoyle, 1960](#)).

Intermediate-Mass Stars: $2 M_{\odot} < M < 8 M_{\odot}$

Massive Stars: $8 M_{\odot} < M < 80 M_{\odot}$

Very Massive Stars: $80 M_{\odot} < M < 300 M_{\odot}$

2.2.3

Relativistic corrections

This analysis has presumed Newtonian mechanics. However, the mass of stars puts them formally into the general relativistic regime, and thus strictly speaking we should treat this as a problem in full GR. In this approach, we would solve the Einstein Field Equations for the interior of the star:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \quad (2.13)$$

with cosmological constant Λ , stress-energy $T_{\mu\nu}$, and the Einstein tensor $G_{\mu\nu}$ given by

²The relationship between a colour-magnitude diagram and the HR diagram is that $\log L = 0.4 (M_{\text{bol},\odot} - M_{\text{bol},*}) \simeq 0.4 (4.74 - M_{v,*} - \text{BC})$, $\log T = \log \left(4600 \left(\frac{1}{0.92(B-V)+1.7} + \frac{1}{0.92(B-V)+0.62} \right) \right)$, where BC is a bolometric correction which depends on the stellar type. This expression for the temperature is a fit provided by [Ballesteros \(2012\)](#).

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \quad (2.14)$$

$$= R^\lambda_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu}R^\sigma_\sigma \quad (2.15)$$

$$= R^\lambda_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu}g^{\sigma\gamma}R_{\sigma\gamma} \quad (2.16)$$

$$= \sum_\lambda R^\lambda_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} R^\alpha_{\sigma\alpha\gamma} \quad (2.17)$$

$$= \sum_\lambda \left(\Gamma^\lambda_{\mu\nu,\lambda} - \Gamma^\lambda_{\mu\lambda,\nu} + \sum_\xi \Gamma^\lambda_{\xi\lambda} \Gamma^\xi_{\mu\nu} - \sum_\xi \Gamma^\lambda_{\xi\nu} \Gamma^\xi_{\mu\lambda} \right) - \frac{1}{2}g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} \left(\Gamma^\alpha_{\sigma\gamma,\alpha} - \Gamma^\alpha_{\sigma\alpha,\gamma} + \sum_\xi \Gamma^\alpha_{\xi\alpha} \Gamma^\xi_{\sigma\gamma} - \sum_\xi \Gamma^\alpha_{\xi\gamma} \Gamma^\xi_{\sigma\alpha} \right), \quad (2.18)$$

where there is still tensorial shorthand:

$$\Gamma^\gamma_{\beta\mu} \equiv \frac{1}{2} \sum_\alpha g^{\alpha\gamma} (g_{\alpha\beta,\mu} + g_{\alpha\mu,\beta} - g_{\beta\mu,\alpha}) \quad (2.19)$$

and the comma-derivative notation is

$$\phi_{,\mu} \equiv \frac{\partial\phi}{\partial x^\mu}, \quad (2.20)$$

such that, for example,

$$g_{\alpha\beta,\mu} = \frac{\partial g_{\alpha\beta}}{\partial x^\mu}.$$

In Equation 2.17, we explicitly reinsert the summation symbols to illustrate the number of simultaneous summations that are taking place implicitly. As one can see, this reasonably simple expression balloons quickly into something intractable by hand and time-consuming to evaluate computationally. Although the metric tensor is generally four-dimensional,³ such as the Friedmann-Lemaître-Robertson-Walker metric (Friedmann, 1922; Lemaître, 1931; Robertson, 1935; Walker, 1937) describing the expansion of the Universe,

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a(t)^2 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix},$$

it often lacks a closed form, especially on large-scale chaotic behaviour that cannot be locally “washed out” (such as how on Universal scales, individual stars can be ignored to describe the homogeneity and isotropy of the Universe), and is obtained via numerical methods.

³And thus the summation indices run from 1 to 4

This is also not considering complications of writing the stress-energy tensor $T_{\mu\nu}$, which may have its own complex form (again, assuming a closed-form solution *exists*) determined via fluid-mechanical processes. All told, there are 64 equations to be solved instead of 4. In addition, the 64 solutions are “worse” to solve: harder to integrate by hand, and requiring greater time complexity. Numerical relativity codes typically simulate one star for the order of kiloseconds of its lifetime (see, for an example, [ud-Doula et al., 2013](#)), and even the most efficient codes take longer than our codes to run. For instance, the PHANTOM 3D MHD code ([Price et al., 2018](#)) can take several days to run high-resolution models. Consequently, for the number of stars we wish to simulate, we choose to adopt a Newtonian framework.

2.3 Supernovae

Pallida mors æquo pulsat pede pauperum tabernas Regumque turres.

*Pale death knocks with the same tempo
upon the huts of the poor and the towers of kings.*

HORACE

A star’s lifetime is, unfortunately, not infinite. The key assumption underpinning stellar evolution is that of hydrostatic equilibrium: the counterplay of gravity against outwards pressure. Whilst this equilibrium holds, the star is stable and resists explosion. It is when this equilibrium is broken – with gravity the dominant force in this scenario – that the star implodes in a core-collapse supernovae (CCSN). From a thermodynamic perspective, as the star evolves, it shrinks as energy is bled off into the interstellar medium. This shrinkage compresses and heats the core, meaning the star’s specific heat capacity is negative. This system is thermodynamically unstable, and thus every star either collapses into a remnant (c.f. black holes, neutron stars) or is sustained in some form by quantum mechanical phenomena (c.f. White Dwarfs, Thorne-Żytkow objects, and other exotica).

In any case, the breaking of hydrostatic equilibrium results generally results in⁴ a supernova, which can be classified into subtypes, as shown in Figure 2.1. In this event, gravity pulls the stellar core inwards, before “bouncing” and ejecting the envelope into the interstellar medium. The physics of this approximately second-long event are highly uncertain, and are in general unable to be modelled by detailed stellar evolution codes. All models we generate – assuming they have not been terminated earlier – will terminate at the moment of supernova.

Importantly for the nucleosynthesis of heavier-than-Iron elements, the supernova shockwave propagating through the core causes a new phase of nucleosynthesis, albeit briefly.

2.4 Two to tango: Binary Evolution

Introducing a second star into the mix complicates the evolution. Whereas before a star losing mass (through, for example, a coronal mass ejection (CME)) was assumed to eject it into the ISM, it may now accrete that

⁴The extent of the veracity of this statement is an area of active research: [Reynolds et al. \(2015\)](#) presents a short recap.

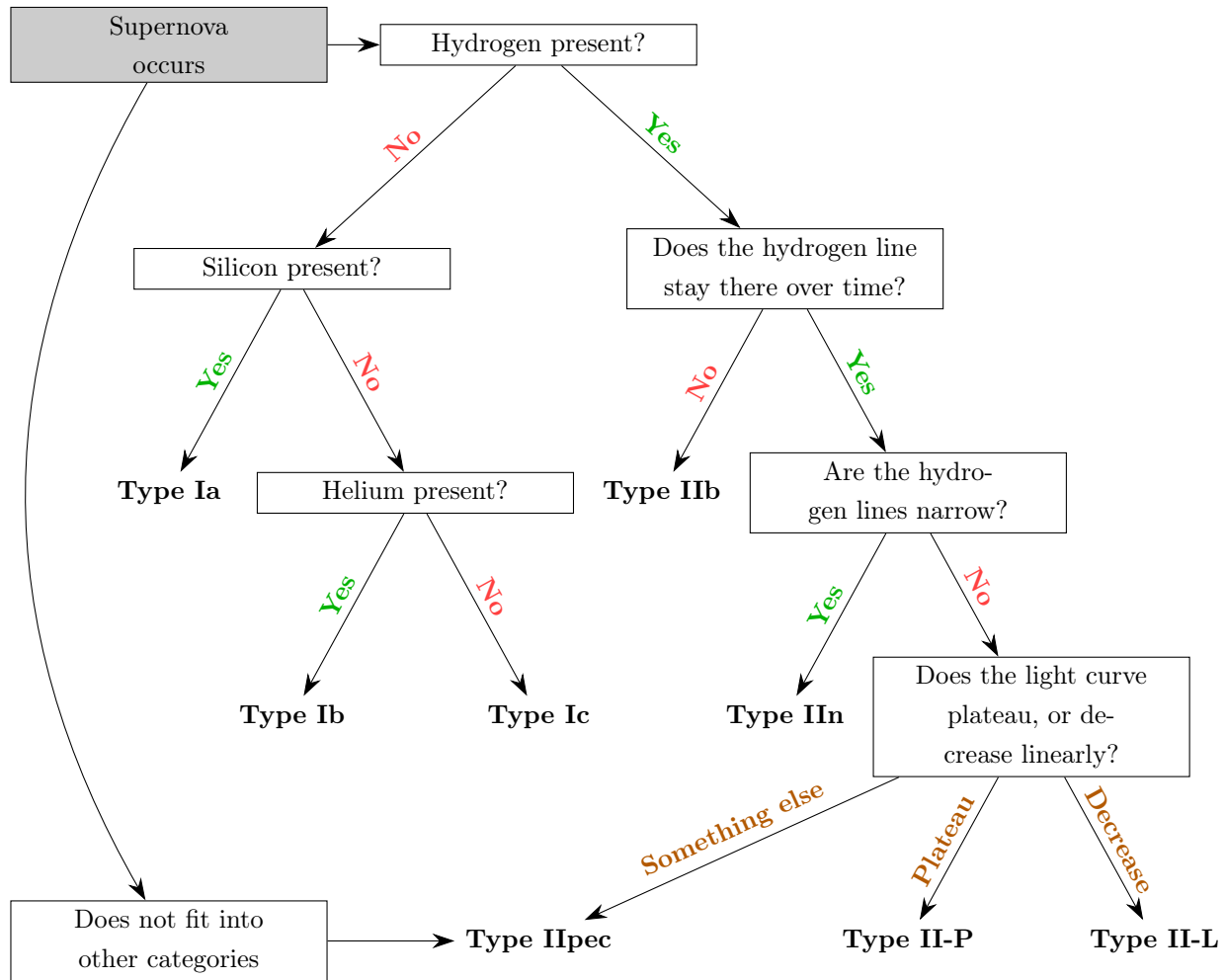


Figure 2.1: A rough outline of supernova classification. One can see that the presence of hydrogen lines distinguishes a Type I from Type II supernova. Type II supernova subtypes are determined by the behaviour of their light curves – how the apparent magnitude changes against time – and thus are determined by the luminosity of the supernova, something detailed models are not able to model. On the other hand, the presence of hydrogen/silicon/helium in the supernova is something that detailed codes *are* able to model.

material directly onto its companion. In addition, either star may in principle increase in radius enough to overlap with its companion, or even touch surfaces – introducing new dynamics into the mix.

2.4.1

Binary star kinematics

There are several formation channels for a binary star system. Binaries may be formed from birth (as discussed in section 2.1, with the caveat that in this scenario two clumps of gas form near each other), via fragmentation of a protostar’s disc, dynamical capture of one star by another (comparable to one of the suggested origins of Phobos and Deimos, per [Burns \(1978\)](#)), or via a higher multiplicity system ejecting one or more of its constituent stars, perhaps via a natal supernova kick.

In any case, once formed and the orbit is stabilised, binary systems obey Kepler’s three laws. In the generalised binary regime, however, the assumption that $M_1 \gg M_2$ (which underpins most statements of Kepler’s laws) is not necessarily true – for instance, the system LSS 3074 has two stellar components with ZAMS masses of approximately $30 M_\odot$ and $10 M_\odot$ ([Raucq et al., 2017](#), and section 3.3 of this thesis). Moreover, the fact that there are two stars in orbit rather than one object orbiting another means that Kepler’s laws must be stated more precisely. Formally, they can be expressed as

1. All orbits are ellipses, with both stars orbiting a common barycentre at the focus of the ellipse,
2. A line segment joining the two stars sweeps out equal areas in equal times,
3. The square of the period of the system is proportional to the cube of the semimajor axis of the ellipse (the binary separation), with the proportionality term dependent on the sum of the masses.

The third law can be stated more precisely as,

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.21)$$

and a schematic of the orbit of a binary system is given in Figure 2.2.

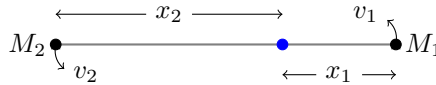


Figure 2.2: A schematic of a simple binary star system. The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a greater distance x_1 to the barycentre by Kepler’s first and Newton’s second laws. Figure adapted from [Richards \(2021\)](#).

2.4.2

Mass Transfer Efficiency

Whilst mass-loss from a single star is effectively 100 per cent lossy – that is, that 100 per cent of the material ejected from the star is lost to the ISM – the same is not necessarily true in a binary. Suppose star A (called the “donor”) with mass M_{donor} grows to exceed its Roche Lobe, the equipotential surface which is tear-drop shaped, with an apex at the inner Lagrangian point $\mathcal{L}_1$. The excess mass will begin to accrete onto star B

(the “accretor”) which initially has mass M_{acc} . Suppose that the mass lost by the donor is δM . After the mass transfer phase is finished, the mass of both stars will be

$$\left. \begin{aligned} M'_{\text{donor}} &= M_{\text{donor}} - \delta M \\ M'_{\text{acc}} &= M_{\text{acc}} + \kappa \delta M \end{aligned} \right\} \quad (2.22)$$

where $\kappa \in [0, 1]$ is the mass transfer efficiency. $\kappa = 0$ corresponds to perfectly lossy – all mass accreted from the donor is lost to the ISM – whereas $\kappa = 1$ is perfect efficiency – all mass accreted from the donor is accepted by the accretor, with none lost to the ISM. The total mass is now

$$M \equiv M'_{\text{donor}} + M'_{\text{acc}} = M_{\text{donor}} + M_{\text{acc}} - (1 - \kappa)\delta M \quad (2.23)$$

Alternatively, we can construct a more complex model of mass transfer, allowing for a variable efficiency. For transfer from star a to b in a binary, this is

$$\dot{M}_a = \frac{M_a - \exp(L_a - \delta L_a)}{dt} - \mathcal{R}_1 M_a + f(a) + \min \left| \mathcal{R}_2 \left(\frac{\log R_a - \log \left(a \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} \ln(1+q^{\frac{1}{3}})} \right) \right)}{2} \right)^3, 10^{-2} \text{M}_{\odot} \text{yr}^{-1} \right|, \quad (2.24)$$

where $f(a)$ is a given mass loss prescription.

2.4.3 Roche Lobes

Algebra's like sheet music. The important thing isn't can you read music, it's can you hear it. Can you hear the music, Robert?

Niels Bohr, IN OPPENHEIMER

A binary system is complicated by the presence of its companions in several different ways. As alluded to in the previous section, the presence of a companion warps space-time in such a way that material can be quite distorted from a sphere or oblate spheroid but still remain gravitationally bound to the system. In the extremal case, a star can grow to fill or even exceed its Roche Lobe, which is the tear-drop equipotential shaped surface within which material remains gravitationally bound to its star. In Figure 2.3 we display these equipotential surfaces for a range of different mass ratios. It is pertinent to note that the shape of all of the Roche Lobes change rapidly at low mass ratios, however at higher mass ratios the shapes change less rapidly. In an equal-mass ratio binary, the binary configuration has rotational symmetry about the $\mathcal{L}_1$ point, and reflectional symmetry about the x -axis and the line $y = \mathcal{L}_1$.

There are two primary Roche Lobe points we consider: the $\mathcal{L}_1$ point and the $\mathcal{L}_2$ point. The $\mathcal{L}_1$ point (also known as the inner Lagrangian point) is the nexus between the two stars, the $\mathcal{L}_2$ point (the outer Lagrangian point) is located on the other side of the secondary to the inner point.

We follow the derivation given in [Rose \(1998\)](#) to find an expression for the Roche potential. First we note that in a rotating reference frame in Cartesian coordinates, we have

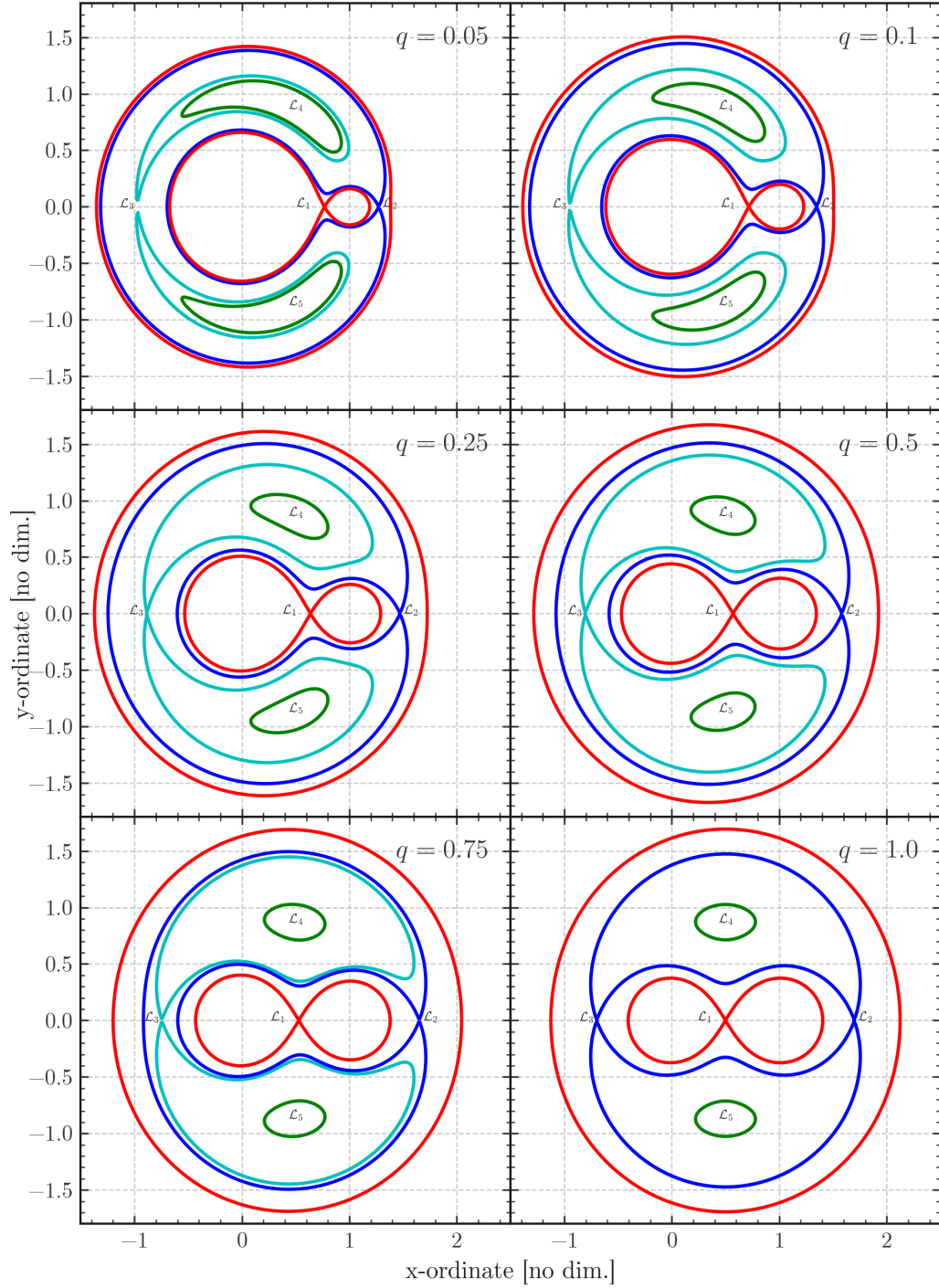


Figure 2.3: The dimensionless Roche potential as a function of mass ratio q . Note that the shape of the Roche potential changes markedly at low mass ratio, and changes more slowly as the mass ratio approaches unity. Distances have been normalised such that the primary star is at $(x, y, z) = (0, 0, 0)$ and the secondary star is at $(x, y, z) = (1, 0, 0)$, i.e., we measure distances in units of separation and that the coordinate system is centered on the primary star.

$$\overbrace{\frac{GM_1M_2}{a^2}\hat{\mathbf{x}}}^{\text{gravitational force}} = \underbrace{-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1)}_{\text{centrifugal force}}, \quad (2.25)$$

where $\hat{\mathbf{x}}$ is the line between the center of mass of both stars, a the separation, $\vec{\omega}$ the angular velocity, and $\vec{r}_1$ the position vector of the primary star. The x-ordinates of the centres of mass are given by

$$\left. \begin{aligned} x_1 &= \frac{M_2}{M_1+M_2}a \\ x_2 &= \frac{-M_1}{M_1+M_2}a \end{aligned} \right\} \quad (2.26)$$

Since we are considering a rotating coordinate system we have that the centrifugal force is given by

$$-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1) = M_1\omega^2 x_1 \quad (2.27)$$

and hence

$$\begin{aligned} \frac{GM_1M_2}{a^2}\hat{\mathbf{x}} &= M_1\omega^2 x_1 \\ \omega^2 &= \frac{GM_1M_2}{M_1x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2}{x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2(M_1+M_2)}{M_2 \cdot a \cdot a^2}\hat{\mathbf{x}} \\ &= \frac{G(M_1+M_2)}{a^3}\hat{\mathbf{x}}. \end{aligned} \quad (2.28)$$

The centrifugal potential of a test mass m at (x, y) is hence

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2) = -\frac{Gm(M_1+M_2)(x^2 + y^2)}{2a^3}, \quad (2.29)$$

and thus its energy is

$$E = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + V(x, y), \quad (2.30)$$

where

$$V(x, y) = -\overbrace{\frac{GmM_1}{\sqrt{(x-x_1)^2 + y^2}}}^{\text{G.P.E. due to primary}} - \overbrace{\frac{GmM_2}{\sqrt{(x-x_2)^2 + y^2}}}^{\text{G.P.E. due to secondary}} - \overbrace{\frac{Gm(M_1+M_2)(x^2 + y^2)}{2a^3}}^{\text{centrifugal potential}}. \quad (2.31)$$

We introduce the dimensionless coordinates measured in units of the separation of the binary (or reduced masses)

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{a}, \quad \xi_1 = \frac{M_2}{M_1+M_2}, \quad \xi_2 = \frac{M_1}{M_1+M_2} \quad (2.32)$$

Our potential now becomes

$$V(\xi, \eta) = -\frac{Gm(M_1 + M_2)}{a} \times \left[\frac{\xi_2}{\sqrt{(\xi - \xi_1)^2 + \eta^2}} + \frac{\xi_1}{\sqrt{(\xi - \xi_2)^2 + \eta^2}} + \frac{1}{2}(\xi^2 + \eta^2) \right]. \quad (2.33)$$

Taking the partial derivative along the ξ -axis (the rescaled x -axis), we get

$$\frac{\partial V}{\partial \xi} = -\frac{Gm(M_1 + M_2)}{a} \times \left[-\frac{\xi_2(\xi - \xi_1)}{\left((\xi - \xi_1)^2 + \eta^2\right)^{\frac{3}{2}}} - \frac{\xi_1(\xi - \xi_2)}{\left((\xi - \xi_2)^2 + \eta^2\right)^{\frac{3}{2}}} + \xi \right] \quad (2.34)$$

Hence the extrema of the potential V are at the points ξ where (taking $\eta = 0$ on the ξ -axis)

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}. \quad (2.35)$$

No analytical function for ξ exists unless either or both of ξ_1, ξ_2 are zero (which would only be the case for a single star), thus we must make approximations. First, we note that if $q = \frac{M_2}{M_1}$ the ξ_1, ξ_2 coefficients can be written as

$$\xi_1 = \frac{q}{1+q}, \quad \xi_2 = \frac{1}{q+1}. \quad (2.36)$$

Define $z = 1 + q$, and then Equation 2.35 becomes

$$0 = \xi - \frac{1}{z} \frac{1}{\left(\xi - \frac{q}{z}\right)^2} - \frac{q}{z} \frac{1}{\left(\xi - \frac{1}{z}\right)^2} \quad (2.37)$$

$$= \xi - \frac{z}{(z\xi - q)^2} - \frac{qz}{(z\xi - 1)^2}, \quad (2.38)$$

which may be rearranged to

$$\xi = \frac{z}{(z\xi - q)^2} + \frac{qz}{(z\xi - 1)^2}. \quad (2.39)$$

With tedious algebra it can be shown that this condition is equivalent to a quintic in ξ with coefficients

$$0 = \sum_{n=0}^5 (-1)^{n+1} a_n \xi^n \text{ where } \begin{cases} a_0 &= q^4 + q^3 + q + 1 \\ a_1 &= 2q^3 + 7q^2 + 6q + 2 \\ a_2 &= 4q^3 + 9q^2 + 6q + 1 \\ a_3 &= q^2 + 2q + 1 \\ a_4 &= 2q^4 + 8q^3 + 12q^2 + 8q + 2 \\ a_5 &= q^4 + 4q^3 + 6q^2 + 4q + 1 \end{cases} \quad (2.40)$$

with solutions given by the eigenvalues of the companion matrix, i.e., ξ obeys

$$0 = \det \begin{pmatrix} -\xi & 0 & 0 & 0 & \frac{q^4+q^3+q+1}{q^4+4q^3+6q^2+4q+1} \\ 1 & -\xi & 0 & 0 & -\frac{2q^3+7q^2+6q+2}{q^4+4q^3+6q^2+4q+1} \\ 0 & 1 & -\xi & 0 & \frac{4q^3+9q^2+6q+1}{q^4+4q^3+6q^2+4q+1} \\ 0 & 0 & 1 & -\xi & -\frac{q^2+2q+1}{q^4+4q^3+6q^2+4q+1} \\ 0 & 0 & 0 & 1 & \frac{2q^4+8q^3+12q^2+8q+2}{q^4+4q^3+6q^2+4q+1} - \xi \end{pmatrix}, \quad (2.41)$$

which may be written more conveniently as

$$0 = \det \begin{pmatrix} -\xi & 0 & 0 & 0 & -2z \\ 1 & -\xi & 0 & 0 & \frac{-2z^3-q^2}{z^4} \\ 0 & 1 & -\xi & 0 & \frac{4q+1}{z^2} \\ 0 & 0 & 1 & -\xi & -\frac{1}{z^2} \\ 0 & 0 & 0 & 1 & 2-\xi \end{pmatrix}. \quad (2.42)$$

This reduces to the quintic

$$\xi^5 - 2\xi^4 + \frac{1}{z^2}\xi^3 - \frac{4q+1}{z^2}\xi^2 - \frac{-2z^3-q^2}{z^4} + 2z = 0. \quad (2.43)$$

Solving either Equation 2.42 or Equation 2.43 admits an exact solution for

Roche Lobe Overflow – the $\mathcal{L}_1$ point

The inner Lagrangian point has the approximation from [Eggleton \(1983\)](#), which in our notation, is

$$\xi = \frac{R_L}{a} = \frac{0.49 \left(\frac{\xi_1}{\xi_2} \right)^{\frac{2}{3}}}{0.6 \left(\frac{\xi_1}{\xi_2} \right)^{\frac{2}{3}} + \ln \left(1 + \left(\frac{\xi_1}{\xi_2} \right)^{\frac{1}{3}} \right)} \quad (2.44)$$

Roche Lobe Overflow – the $\mathcal{L}_2$ point

2.4.4

Common Envelope Evolution

Now we turn to one of the highly uncertain phases of binary evolution. If one star's radius R_1 grows enough such that the radius is greater than the separation between the two binaries, i.e., $R_1 > a$, we say that the star has entered Common Envelope Evolution (CEE). There are several effects that occur here. Firstly, due to orbital drag from the donor star (*1), the recipient star (*2) will shrink in orbit, i.e., the stars will come closer together. The relationship between the initial separation and the final separation is debated, and here we review some of the most common prescriptions.

First, the condition for CEE to occur is that $R_1 > a$. For a star to reach this radius its Roche Lobe, denoted R_L , must also be greater than this radius.

The formula approximating the Roche Lobe (to 1% error in mass ratio q) is, per [Eggleton \(1983\)](#),

$$\frac{R_L}{a} = \frac{R_L}{a} = \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} + \ln\left(1 + q^{\frac{1}{3}}\right)} \quad (2.45)$$

Note that the condition that $R_1 > a$ can be manipulated to $R_L > a$, or $\frac{R_L}{a} > 1$. This is true over the reals for $0.21 < q < 0.29$. Note that this q is not the *initial* (ZAMS) mass ratio, but is instead the mass ratio at a given instant. Moreover, this is not a sufficient condition for CEE to occur (that is, we are not claiming that if this mass ratio is met, the star must be undergoing CEE), but rather a necessary condition for when CEE is even possible in the first instance.

The $\alpha - \lambda$ prescription

The most common CEE prescription is the $\alpha - \lambda$ prescription, which takes

$$\frac{m_1 m_{1,\text{env}}}{\lambda R_1} = \alpha_{\text{CE}} \left(-\frac{Gm_1 m_2}{2a_i} + \frac{Gm_{1,\text{c}} m_2}{2a_f} \right) \quad (2.46)$$

where a subscript “c” denotes a core mass and a subscript “env” denotes an envelope mass. There are apparently two free parameters: α_{CE} , the fraction of the orbital energy used to eject the envelope (or “common envelope efficiency” parameter):

$$\Delta E_{\text{orb}} = \alpha_{\text{CE}} E_{\text{bind}} \quad (2.47)$$

and λ , a structure constant related to the density distribution ([De Kool, 1990](#)). λ is a local parameter – dependent on the structure of the star – and α_{CE} is commonly taken as a global parameter ~ 1 ([Ivanova et al., 2013](#)), though [Scherbak & Fuller \(2023\)](#) has recently suggested $\alpha_{\text{CE}} \sim \frac{1}{3}$ is more appropriate. It is common, though unphysical, to take the combined parameter $\alpha_{\text{CE}}\lambda$ as the *global* free parameter. We will in general not follow this approach; as defined, λ depends on the stellar structure and the stellar binding energy. Hence, assuming it free to vary let alone able to be globally affixed is not a logical choice other than to save computational time.

The γ prescription

The γ prescription arises as a consideration of conservation of angular momentum. Let J be the orbital angular momentum of the binary. Then the γ prescription takes

$$\frac{\Delta J_{\text{lost}}}{J_i} = \frac{J_i - J_f}{J_i} = \gamma \frac{m_1 - m_{1,\text{c}}}{m_1 + m_2}. \quad (2.48)$$

Once an expression for J_f is found, one can transform it into the period via

$$a_f = \frac{m_1 m_2 \sqrt{1 - e^2} P_0}{J_f}, \quad (2.49)$$

where masses are in solar masses, the orbital period is in years, and the semi-major axis is in AU ([Eggen, 1963](#)).

Consider an extremal case where a star composed entirely of helium exists in a binary with a main sequence star. The simple computational regime of taking mass from one star and adding it to the second breaks down, as this would reduce the amount of one element in the binary and increase the amount of the secondary.

Wit nodded.

“Oh, is this the place where I talk?” Kaladin said.

“If you wish.”

“I don’t wish. Get on with the story.”

Rhythm of War, BY BRANDON SANDERSON

After a supernova, the star leaves behind its core as a compact remnant. Generally, these are either white dwarves, black holes, or neutron stars. The interior structure of each is either complicated or, in the case of black holes, impossible to observationally probe without violating causality. Their orbital motions are now deterministic from four initial parameters: m_1, m_2, a , and e . Motion is now wholly determined by the emission of gravitational radiation through (detectable) gravitational waves. The pertinent equations of motion are given by [Peters \(1964\)](#), namely,

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right) \quad (2.50)$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304} e^3 \right) \quad (2.51)$$

with the constant β defined as

$$\beta \equiv \frac{64}{5} \frac{G^3 m M (m + M)}{c^5}. \quad (2.52)$$

Now if one defines the constant c_0 through

$$c_0 = a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304} e_0^2 \right)^{-\frac{870}{2299}}, \quad (2.53)$$

then the time from the second supernova to the conclusion of the inspiral, termed the *coalescence time*, will be given by

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2 \right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de. \quad (2.54)$$

The integral has no clean closed-form solution, and a numerical integrator is typically used to compute the coalescence time. However, some approximations such as [Mandel \(2021\)](#) do exist and approximate the coalescence time rather well over the entire range of mass ratios.

The merger rate of a population of compact binaries at time t after formation can be formalised via the heuristic

$$m(t) = \# \text{ of binaries for which } T_c + T_{\text{rej}} + T_{\text{evo}} < t, \quad (2.55)$$

where T_c is the coalescence time, T_{rej} is the rejuvenation age (the age at which the primary star last accreted mass), and T_{evo} is the evolution time-scale (the length of time the primary spends on the main sequence). $m(t)$ is called the Delay-Time Distribution (DTD) of a population.

However, the underlying assumption in this is that the population was formed after an instantaneous burst of star-forming activity at time $t = 0$, which is clearly unphysical. We must account for the so-called Star Formation Rate Density (SFRD) $\psi(z)$, assumed to be taken from [Madau & Dickinson \(2014\)](#):

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \text{ M}_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (2.56)$$

where z is the redshift. For the purposes of converting between lookback time t and redshift z , we assume a standard LCDM cosmology with $\Omega_M = 0.3, \Omega_\Lambda = 0.7, \Omega_k = 0, H_0 = 69.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Of course, this ignores the effect of mixed-metallicity models, which we account for using the prescription of [Langer & Norman \(2006\)](#), whereupon the SFRD is multiplied by a factor χ :

$$\chi(z, Z) = \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)}, \quad (2.57)$$

where $\hat{\Gamma}$ and Γ are the upper incomplete and complete Gamma functions respectively. This means our Stellar Formation History (SFH) – which takes into account both redshift and metallicity – denoted $\Psi(z, Z)$, is given by

$$\Psi(z, Z) = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)} dz, \quad (2.58)$$

where z_1 and z_2 are the redshifts of our bin edges. The event rate as a function of time, $\mathcal{R}(t)$, is therefore given by

$$\mathcal{R}(t) = \Psi(z_t) * m(t), \quad (2.59)$$

where z_t is the redshift corresponding to lookback time t . Detail on how this convolution is computed may be found in [Richards \(2021\)](#).

A star is not a planet – there is no firm boundary demarcating the inner of a star from the outer. The common definition of the “surface” of a star is the surface of last scattering (photosphere), which is the point radially in which the mean free path of a photon emitted from the core of the star becomes meaningfully

long (internally, $\ell_\gamma \sim 2$ cm, externally, $\ell_\gamma \sim 4.4$ pc).⁵ However, outflows from the star as well as the potential interaction of a partner in a common-envelope event mean that this surface is not necessarily firm: there is no sharp turn from stellar material to empty space. Thus, the simulation of stellar atmospheres is of interest to the stellar astrophysicist. However, this introduces another rather large layer of complexity. The synthesis of a stellar spectra, for example, is not a trivial task, as well as simulating the matter content. This thesis is concerned with detailed modelling of the stellar *interior*, and thus the atmosphere is an interesting sidenote, but one we shall not pursue.

2.7 Numerical implementation

I think there is a world market for maybe five computers.

THOMAS WATSON, PRESIDENT OF IBM, 1943

In this section, we provide an overview of the four codes we either use extensively or have developed to carry out the work of this thesis. These four codes are:

1. The Cambridge STARS code (subsection 2.7.1), developed by Peter Eggleton, which we use to solve the equations of stellar structure and evolution,
2. The TUI code (subsection 2.7.2), which performs our binary population synthesis,
3. The KAITIAKI code (section 3.1), which is our grid-manager code and allows modification of the input parameters to the Cambridge STARS code programmatically, and
4. The TAKAHE code (subsection 4.1.3), which is our binary compact remnant evolution code.

2.7.1 The Cambridge STARS code

Brief History

*And through the night, behind the wheel, the mileage clicking west
I think upon Mackenzie, David Thompson and the rest
Who cracked the mountain ramparts and did show a path for me
To race the roaring Fraser to the sea.*

Northwest Passage, BY STAN ROGERS

Stellar modelling became a field of active research in the 1950s. Prior to the invention of the computer, a model was treated as a mathematical process and thus was computed – often labouriously – by hand. For instance, in [Eggleton \(1967\)](#), Eggleton writes that

⁵The mean free path is a function of the particle under consideration: for a neutrino, internally $\ell_\nu \sim 3.68$ pc and externally $\ell_\nu \sim 81$ Rpc $\sim 3 \times 10^{18}$ times the diameter of the observable Universe.

Since we have seen in Section 3 that the structure of the flame zone can be obtained independently (almost) of the rest of the star, it is reasonable to expect that instabilities of the type considered by Schwarzschild & Härm can also be examined in this way, so that the considerable labour of computing the structure of an entire star can be avoided.

– Eggleton (1967)

The Cambridge STARS code (hereafter simply STARS) was one of the first detailed stellar evolution codes to be written, though initially it only modelled one star in detail. To model a binary system, tweaks were made whereupon the primary would be modelled in detail whilst approximating the secondary with the equations of Hurley et al. (2002) until the primary supernova, at which point the secondary would be evolved in detail. This had its merits; notably computation time was saved, and the results were generally accurate anyway.

However, Stancliffe & Eldridge (2009) introduced detailed modelling of *both* stars in a binary system into the STARS code – that is, the interiors of both stars were computed in detail, alongside the simultaneous computation of the orbital parameters. Since then, a wide array of improvements have been made to the numerical stability of the code, as well as introducing or refining new physics.

Matrix Inversion

The heart of STARS is a matrix inversion algorithm. Although computing the inverse of a matrix is a polynomial time process, it can still take considerable time to invert. STARS implements a Henyey matrix (Henyey et al., 1964), which is of size $n = (4 + n_{\text{comp}})N_{\text{mesh}}$, with n_{comp} the number of equations to be solved and N_{mesh} the number of meshpoints in the model. An individual element of the Henyey matrix is given by

$$\mathbf{H}_{ij} = \frac{\partial (\text{equation } i)}{\partial (\text{variable } j)}. \quad (2.60)$$

For a “typical” binary computation, with $n_{\text{comp}} = 30$ equations and $N_{\text{mesh}} = 199$ meshpoints, the matrix is already 6766×6766 , for a binary which evolves up the AGB (the theoretically most complex calculation we perform) with $N_{\text{mesh}} = 999$, the matrix is 33966×33966 . Introducing k new equations into the code causes the runtime of the matrix inversion calculation to scale like $\mathcal{O}(n^3 + k^3 N_{\text{mesh}}^3)$.

The matrix inversion is simple Gauss-Jordan inversion, which is of complexity class (for an $n \times n$ matrix) $\mathcal{O}(n^3)$. Coppersmith & Winograd (1987) showed that this could be optimised to $\mathcal{O}(n^{2.376})$ and conjectured reducibility to $\mathcal{O}(n^2)$. Since our code is embarrassingly parallelisable, and can be executed in parallel simply by increasing the number of cores available, the payoff for decreasing the runtime by one polynomial order is inconsequential, and thus we use the straightforward implementation.

Initial Condition Selection

In this section we provide an account for the selection of our general initial conditions, unless otherwise noted in this thesis.

In our implementation of Mixing Length Theory (MLT) (Vitense, 1953; Böhm-Vitense, 1958; Cox & Giuli, 1968), we use an MLT parameter of $\alpha_{\text{MLT}} = 1.985$, selected via calibration against a model of the Sun. This is effectively a dissipativity parameter, and represents the depth at which a parcel of matter penetrates into a lower layer before becoming fully dissipated. The value was selected by modelling a $1 M_{\odot}$ star and varying

only the mixing length parameter until it had a luminosity of $1 L_{\odot}$ and a radius of $1 R_{\odot}$, at the present age of the Sun.

Timestep control

Computational considerations

Numerical instabilities

The TUI code

TUI is the core population synthesis code within the BPASS project. TUI takes as input the output from the STARS code.

Chapter Summary

Looking up, he saw a thousand stars glittering in the deep velvet of a night with no moon. He knew them all, their stories and their names. He knew them in a familiar way, the way he knew his own hands.

The Name of the Wind, BY PATRICK ROTHFUSS

Stellar Evolution Synthesis: Results

*Begin at the beginning, the King said, very gravely, and go on till
you come to the end: then stop.*

Alice's Adventures in Wonderland, BY LEWIS CARROLL

As discussed in section 2.4, the presence of a companion measurably alters the evolution of a star.

This chapter is structured as follows. First, we will discuss the grid-watcher code we have developed to both manage the STARS code, namely KAITIAKI. We will then briefly explore some of the general features of binary evolution before moving into two areas of particular interest: the close-contact regime (where the binary period can be as short as 85 min), and variable composition accretion, where material being accreted is allowed to vary.

3.1 The KAITIAKI grid-watcher code

3.2 General Binary Evolution

Captain Holt: *It works! I've never been more proud of you for
anything in your life.*

Terry: *I mean, I've solved a lot of cases for you.*

Captain Holt: *And yet crime has continued.*

BROOKLYN NINE-NINE

3.2.1 Rationale

In Figure 3.2 we display in three subplots the evolution of an $8 M_{\odot}$ single star, a $10 M_{\odot}$ single star, and an $8 + 10 M_{\odot}$ binary with a period of 30 d. The code to run this binary model is given in Code Sample A.3.

3.2.2 Fiducial comparison

3.3 Close-Contact Binaries

*All hands, this is the Captain. Begin prepping all stations for first
contact.*

CAPTAIN ED MERCER, "THE ORVILLE"

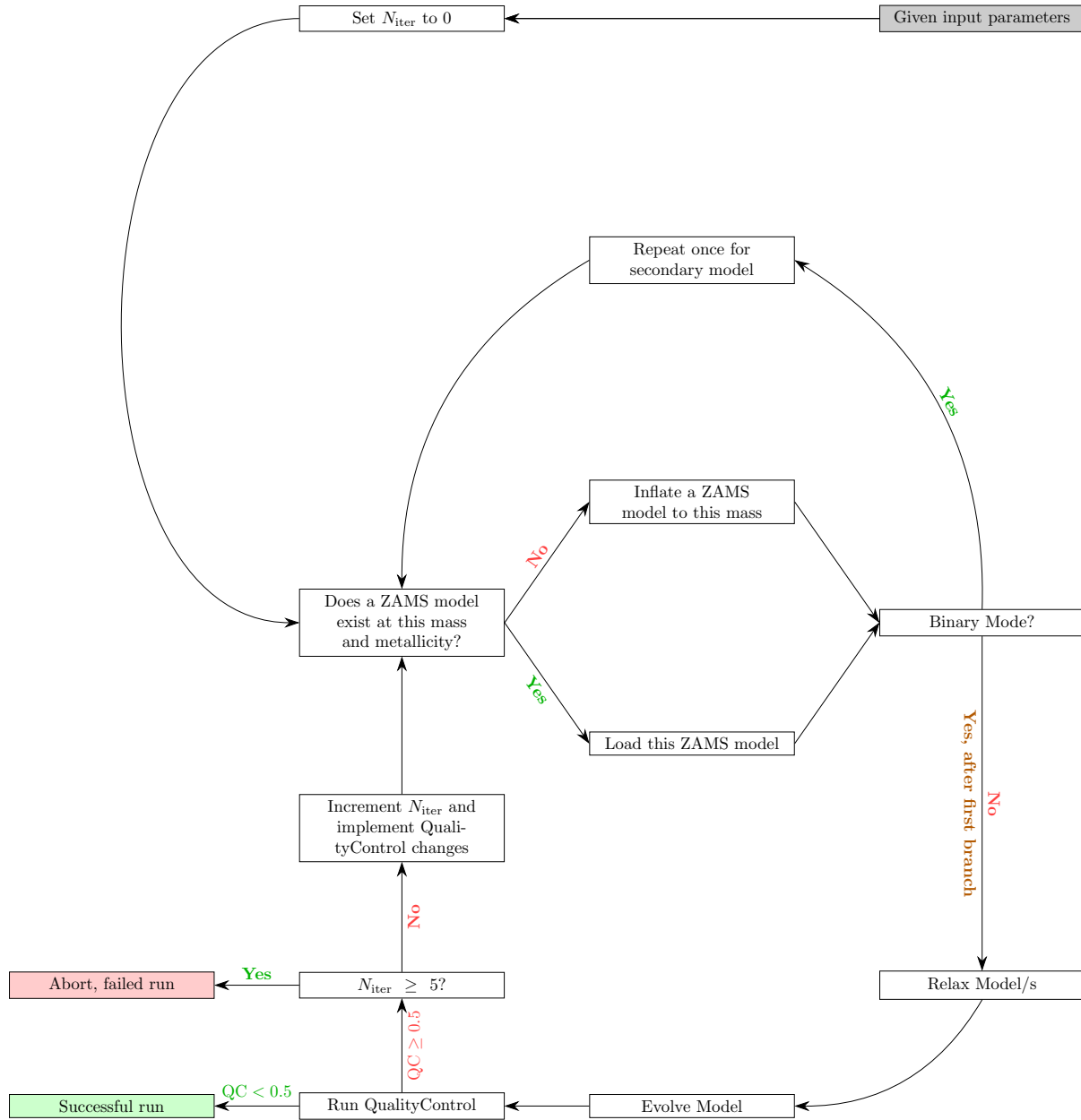


Figure 3.1: Control flow of a grid calculation performed by KAITIAKI.

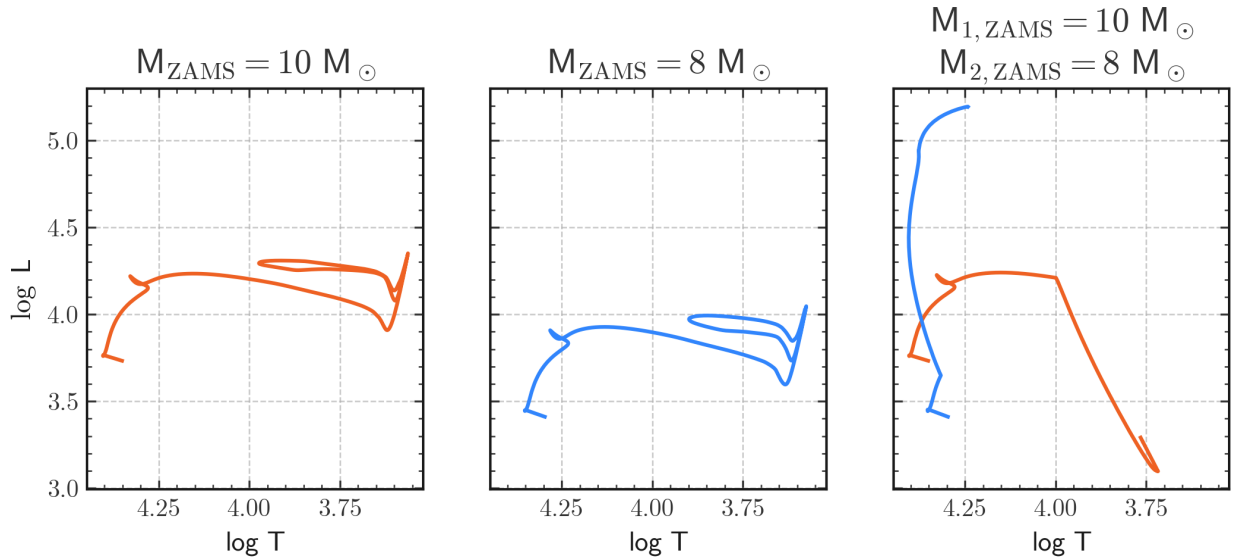


Figure 3.2: The evolution of an $10 M_{\odot}$ (leftmost) and $8 M_{\odot}$ (centermost) star as both single stars and as a binary with a period of 30 d (rightmost). Roche lobe overflow occurs for the primary when it begins to cross the Hertzsprung gap, and for the secondary shortly off the main sequence.

3.3.1 Rationale

3.3.2 Parameter Selection

3.3.3 Interaction Effects and Roche Lobe overflow

3.4 Stripped Helium Stars

3.5 Chapter Summary

Post-Supernova Kinematics: The Natal Supernova Kick

*I fear not the man who has practiced 10,000 kicks once,
but I fear the man who has practiced one kick 10,000 times.*

BRUCE LEE

If you torture the data enough, they will confess.

SALVATORE VITALE

4.1

Overview

The terminus of a massive star's life is a core-collapse supernova, where the hydrostatic equilibrium between the pressure of the star and its self-gravity is broken, and the star collapses in on itself. Once the star reaches its minimum radius, it rebounds, ejecting an amount of mass we denote M_{ej} from the star and leaving a remnant of mass M_{rem} behind. If the ejection is asymmetric, this endows the nascent remnant with a velocity v_{k} opposite to the more massive ejecta, which is termed a natal supernova kick.

As a consequence of conservation of momentum, it can be shown that if the ejecta is assumed to eject along a preferred axis, the velocity of the kick can be written as

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta \left(\frac{M_{\text{NS}}}{M_{\text{rem}}} \right), \quad (4.1)$$

for free parameters α and β (both measured in km s^{-1}). Here, M_{NS} is the mass of a Neutron Star (NS); if the kick is applied to a NS, then $M_{\text{NS}} = M_{\text{rem}}$ and the coefficient reduces to unity. If the kick is applied to a Black Hole (BH), then we can take $M_{\text{NS}} = 1.4 M_{\odot}$, which serves to convert the distribution from one in velocity-space to one in momentum-space; see [Bray & Eldridge \(2016, 2018\)](#); [Richards et al. \(2023\)](#) for further detail.

This parameterization introduces, effectively, four free parameters in $\alpha, \beta, \theta, \phi$ (the latter two being azimuthal and equatorial angles of the ejecta mass). We assume that the preferential axis is sampled uniformly across the 4π steradians of the star, and that our only two free parameters are α and β . It is pertinent to note two things due to this assumption: firstly, we are not assuming that the ejection occurs along one axis and that the kick is aligned along a different, random, axis. Rather, we are assuming that the ejection occurs randomly and that the kick is antialigned to that. The second note, explored further in subsection 4.1.2, is that there is considerable uncertainty around the nature of the kick and the precise mechanism that causes it to eject along a given axis. This is the reason we sample the kick direction uniformly, though we will simulate sufficiently many systems (on the order of $\mathcal{O}(10^{12})$) that this will be marginalised over.

The “traditional” way to refine the two free parameters, as outlined in [Bray & Eldridge \(2016, 2018\)](#), is to refine based on isolated pulsar velocities. This methodology, however, has significant drawbacks. Notably, the reliance on one observation can engender biases in the result at the expense of no longer predicting accurate results for other observables. For instance, in the limit of ultra-stripped supernovae (USSNe) where the ejecta-remnant mass ratio becomes small, the fiducial kick in [Bray & Eldridge \(2018\)](#) would produce kick velocities that are strongly negative, contrasted with the weak kicks we expect from these supernovae (e.g., [Willcox et al., 2021](#)). It becomes evident that a new pipeline is needed, whereupon more constraints than simply pulsar velocities are tested. In this Chapter, we outline four such tests that form the basis of the work performed in [Richards et al. \(2023\)](#).

4.1.1

Parameter space

We perform our computation on three populations, outlined in Table 4.1. These populations simulate 1,000 kicks per system sampled. The aim of the three populations is twofold. Firstly, by generating FINEGRID and WIDEGRID, we are able to look both at fine detail in the region at which the most likely value is expected to lie and also the large-scale behaviour over wider parameter ranges. Secondly, we use a Hobbs distribution as a fiducial comparison, discussed below.

Population Name	α_{low} [km s ⁻¹]	α_{high} [km s ⁻¹]	β_{low} [km s ⁻¹]	β_{high} [km s ⁻¹]	Step Size [km s ⁻¹]
FINEGRID	0	400	-200	200	10
WIDEGRID	0	700	-350	350	50
HOBBS1000	-	-	-	-	-

Table 4.1: The populations under study. HOBBS1000 is a fiducial kick prescription as given by [Hobbs et al. \(2005\)](#), which is a Maxwellian distribution with distribution parameter $\sigma = 265 \text{ km s}^{-1}$, based on an empirical fit to the velocities of 233 pulsars. All three prescriptions simulated 1,000 kicks for each system, with the resultant kick being the mean velocity sampled, and the direction of the kick being sampled uniformly across the 4π steradians of the orbit.

4.1.2

Natal Kick Prescriptions

It is pertinent to note that there are conflicting accounts as to the cause of the kick (see, for example, [Fryer et al., 1998](#); [Arzoumanian et al., 2002](#); [Bombaci & Popov, 2004](#); [Hobbs et al., 2005](#); [Bray & Eldridge, 2016](#); [Verbunt et al., 2017](#)), there are conflicting accounts as to the form and distribution of the kick too, as discussed in, for example, [Blaauw \(1961\)](#); [Gunn & Ostriker \(1970\)](#); [De Loore et al. \(1975\)](#); [Sutantyo \(1978\)](#); [Hobbs et al. \(2005\)](#); [Baker et al. \(2008\)](#); [Wongwathanarat et al. \(2013\)](#); [Bray & Eldridge \(2018\)](#); [Giacobbo & Mapelli \(2020\)](#); [Mandel & Müller \(2020\)](#).

We stress that we are not attempting to prescribe *which* prescription is somehow “best”, but seek an easy-to-configure kick which allows us to configure kicks without changing any of the physics of how our population was formed. This can be done by modifying the kick of [Bray & Eldridge \(2016\)](#), which takes

two free parameters which can be changed without imposing constraints on physics. This prescription is hereinafter referred to as the “Bray kick”. Similarly, we require for the purposes of computing log-Likelihoods in section 4.3 a comparison kick, for which we select the kick of [Hobbs et al. \(2005\)](#), a well-known empirically derived kick. This prescription is hereinafter referred to as the “Hobbs kick”.

The Bray kick velocity is prescribed by Equation 4.1. The coefficient to the β term, $\frac{M_{\text{NS}}}{M_{\text{rem}}}$, serves to generalise the kick from purely NSs to BHs as well. When applied to a BH, M_{NS} is taken to be $1.4 M_{\odot}$, and conversely when applied to a NS, M_{NS} is by definition the remnant mass, so the kick velocity becomes

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta. \quad (4.2)$$

On the other hand, the Hobbs kick velocity is given by the Probability Density Function (PDF)

$$P(v_{\text{k}}) dv_{\text{k}} = \sqrt{\frac{2}{\pi}} \frac{v_{\text{k}}^2}{\sigma^3} \exp \left(-\frac{v_{\text{k}}^2}{2\sigma^2} \right) dv_{\text{k}}, \quad (4.3)$$

where $\sigma = 265 \text{ km s}^{-1}$ is the distribution parameter found by fit in [Hobbs et al. \(2005\)](#) to the velocities of 233 pulsars.

4.1.3

The TAKAHE binary evolution code

As part of this work, we have developed the TAKAHE binary evolution code, which solves the coupled equations of motion Equation 2.50 and Equation 2.51. Originally, to facilitate the equations being solved rapidly, we wrote the code as a PYTHON wrapper to a JULIA ([Bezanson et al., 2017](#)) integrator. However, the PYJULIA library ([Arakaki et al., 2023](#)) proved to be fragile and is not in fact supported officially on recent Python versions. To facilitate the potential interoperability between KAITIAKI and TAKAHE (and to remove an obnoxious loading time when the JULIA runtime was loaded), we elected to rewrite the core integrator into PYTHON and to optimise it using traditional PYTHON methods instead.

There are two relevant functions in TAKAHE. The remaining functions are utility functions that allow us to (for instance) convert between lookback time / redshift, and period / separation. These functions are `integrate_eoms()` (Code Sample A.1), which solves the coupled differential equations for $\frac{da}{dt}$ and $\frac{de}{dt}$ (Equation 2.50 and Equation 2.51), and `integrate_timescale()` (Code Sample A.2), which solves Equation 2.54 for the coalescence time of the binary. With sufficient performance optimisations,¹ `integrate_timescale()` is able to achieve speeds of approximately 18,000 numerical integrations per second on an office PC on a sample of Neutron Star - Black Hole Binary (NS-BH) binaries in the LISA dataset ([Tang, 2024](#)).

Algorithmically, both codes employ a simple Euler integration routine (where $f_{n+1} = f_n + \frac{df_n}{dt} dt$) and the choices work as follows:

1. When `integrate_eoms()` takes a step in time, it evaluates da and de . It then computes the relative changes in both e and a that arise by taking this timestep (through $\delta f = \left| 0.01 \frac{f}{f} \right|$), and takes a timestep corresponding to the smallest change between a and e , or half of the width of the smallest BPASS time bin.
2. `integrate_timescale()` breaks up Equation 2.54 into four cases dependent on the initial eccentricity. Let τ be the coalescence time of a circular binary ($\tau = \frac{a_0^4}{4\beta}$), then the cases are:

¹Primarily, just-in-time compilation with the NUMBA ([Lam et al., 2015](#)) library.

$$t_c = \begin{cases} \tau & e_0 = 0 \\ \frac{c_0^4}{4\beta} e_0^{\frac{48}{19}} & e_0 < 0.01 \\ \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} (1 + \frac{121}{304} e^2)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de & 0.01 \leq e_0 \leq 0.99 \\ \tau \frac{768}{425} (1 - e_0^2)^{\frac{7}{2}} & 0.99 < e_0 \end{cases}, \quad (4.4)$$

where all symbols have the same meaning as in section 2.5. The integral in the intermediate case is evaluated numerically, taking a fixed timestep of $de = \frac{e_0}{10000}$ such that the discretised sum becomes nothing more than

$$T(e_0) = \frac{e_0}{10000} \sum_{e=0}^{e_0} \frac{e^{\frac{22}{19}} (1 + \frac{121}{304} e^2)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}}, \quad (4.5)$$

and dependence of the coalescence time t_c on the initial separation a_0 is handled through the c_0 term. Since the step size is an arbitrary choice, a tradeoff can be made between accuracy and runtime. Figure 4.1 shows the effect of changing the step size ($\frac{e_0}{10000}$ in Equation 4.5) on convergence to a final result.

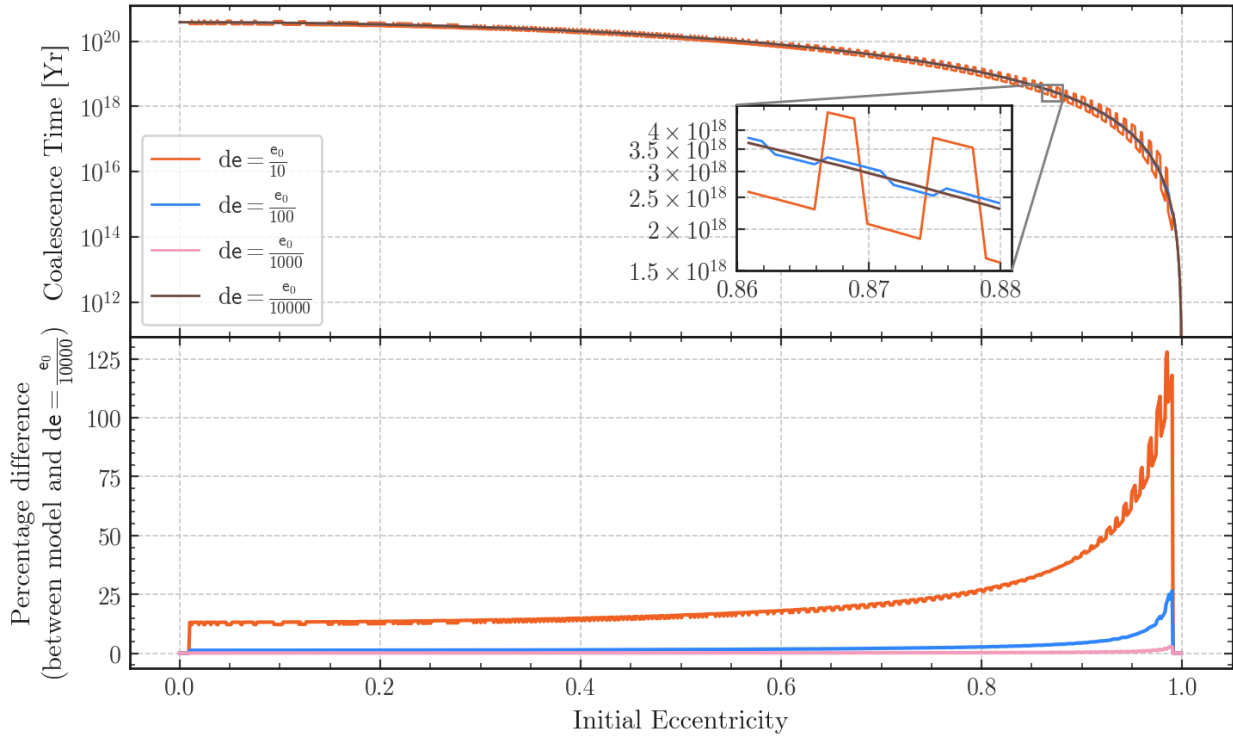


Figure 4.1: The effect of varying de in Equation 4.5 on the coalescence time. Although hard to see, the pink line does oscillate above and below the brown line. This binary was assumed to have $m_1 = m_2 = 1.44 M_\odot$ and $a_0 = 1970.5 R_\odot$ (the mean separation of a sample of binaries with $\alpha = 100 \text{ km s}^{-1}$, $\beta = -170 \text{ km s}^{-1}$).

4.2 Merger Rates of BNS systems

The first test is to match the observed merger rates of Binary Neutron Star (BNS) systems. To do this, we use the LVK Collaboration catalogue of results from Operating Run 3 (O3). The simplest test is to match the merger rate today, which we denote as $\mathcal{R}_0$, to that observed by LVK Collaboration (LIGO-Virgo, 2021; LIGO-Virgo-KAGRA, 2021). We display this in Figure 4.2 alongside the previously reported fiducial values of $\alpha = 100 \text{ km s}^{-1}$ and $\beta = -170 \text{ km s}^{-1}$, the theoretical merger rate calculation performed by Eldridge et al. (2019), and the observed merger rates from LIGO-Virgo (2021) and LIGO-Virgo-KAGRA (2021). Of note, we term the contour defined by LIGO-Virgo (2021) the “LIGO O3 contour”.

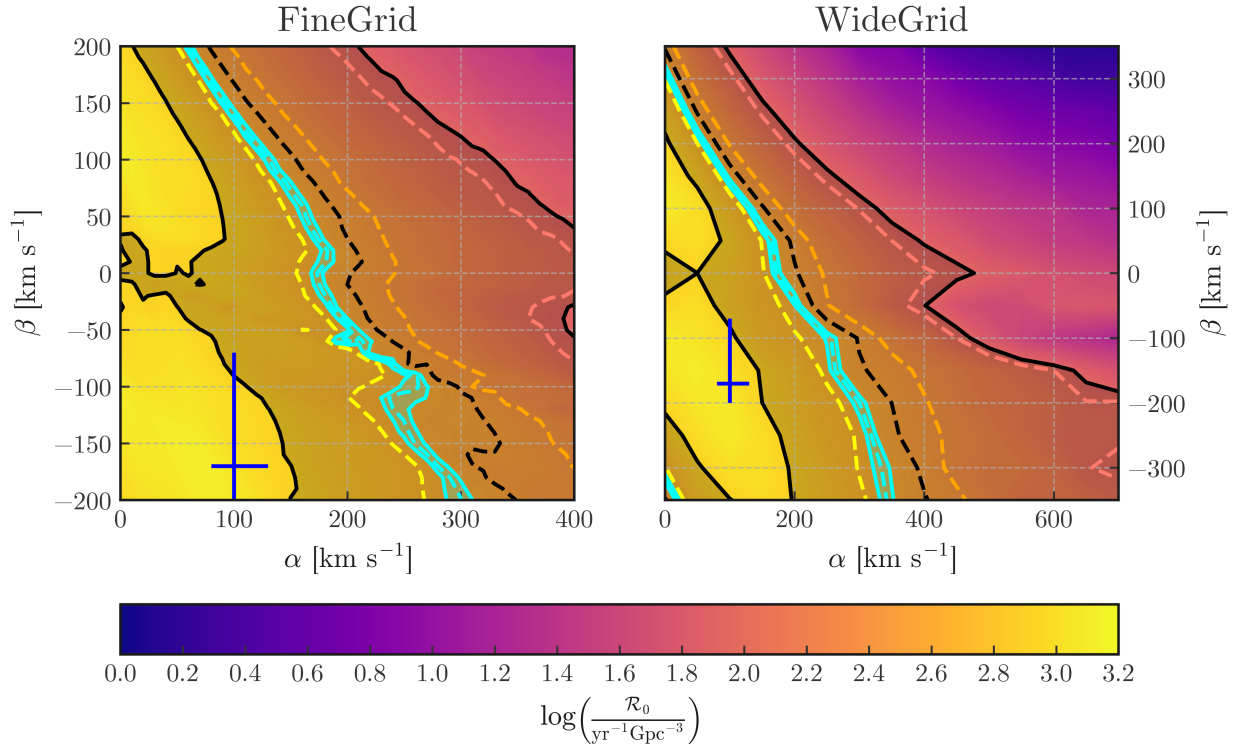


Figure 4.2: The merger rate as a function of kick parameters α and β . The cyan contour is the theoretical merger rate as reported by Eldridge et al. (2019). The black contour is the merger rate observed in LIGO-Virgo (2021). The remaining contours are the merger rates reported in LIGO-Virgo-KAGRA (2021) (pink: BGP, yellow: MS, orange: PDB [ind], see LIGO-Virgo-KAGRA (2021) for an explanation of each model). Shaded regions represent uncertainties, and uncertainties are not reported for the LIGO-Virgo-KAGRA (2021) results as they would cover the entire plane. The blue cross is the fiducial value of α and β from Bray & Eldridge (2018). This value lies off the previous work’s contours as the rejuvenation age calculation has been improved in TUI in preparation for the next major BPASS release.

Period-Eccentricity Distributions

Since, in the absence of any mass transfer, the motion of two compact binaries is wholly dependent on their initial separation and orbital eccentricity, they trace out paths in a two-dimensional phase space. In general, since binaries circularise over time, and by the emission of gravitational waves their orbits shrink, a binary will start in the upper-right region of period-eccentricity space, and evolve towards the bottom left. As a result, given a population of binary stars, we can design a density plot, where the density is effectively the likelihood of a binary selected at random having the parameters we sample.

Log-Likelihood calculations

Computationally, for each binary in our sample, we trace its path through period-eccentricity space, and at every timestep we add $w \times dt$ to the density plot, where dt is the timestep, and w is the weight term output from the TUI code proportional to the IMF weight (where the constant of proportionality is related to the uniformly sampled kick direction). The result of this computation, for the parameter values that lie along the LIGO O3 contour, are shown in Appendix B. Given a catalogue of N (observed) binary stars, we can then compute the likelihood of the model as

$$\ln \mathcal{L} = \sum_{n=0}^N \ln p(\ln P_n, e_n), \quad (4.6)$$

where $p(\ln P_n, e_n)$ is the content of the density plot at the location of the n^{th} observed BNS system. Then, the Bayes factor (defined as the likelihood ratio) can be computed as

$$\mathcal{K} \equiv \frac{\mathcal{L}}{\mathcal{L}_{\text{Hobbs}}} \implies \ln \mathcal{K} = \ln \mathcal{L} - \ln \mathcal{L}_{\text{Hobbs}}, \quad (4.7)$$

where $\mathcal{L}_{\text{Hobbs}}$ is the log-Likelihood of the Hobbs kick. We use a catalogue of fourteen BNS systems originally compiled in [Vigna-Gómez et al. \(2018\)](#), shown in Table 4.2. From this catalogue, we sample the period-eccentricity distribution at that space and take that as the probability of observing that binary, thus allowing us to use the content of the period-eccentricity distribution as the $\ln p$ term in our summand of Equation 4.6.

Single-Star Kick Velocities

The traditional methodology for refining a kick prescription is to consider the velocity distribution of isolated pulsars. Thus, given a suitable observational dataset, we would be able to compute the expected kick distribution and compare it, via a Kolmogorov-Smirnov (K-S) test, to the observed data set. The K-S statistic between distributions $\mathcal{F}$ (the observed distribution) and $\mathcal{G}$ (the compared distribution) is given by

$$\text{KS} = \max |\text{cdf}(\mathcal{F}) - \text{cdf}(\mathcal{G})|, \quad (4.8)$$

where cdf means the Cumulative Distribution Function (CDF) function, defined as the integral of the PDF. Given the CDF is bounded between 0 and 1 inclusive, the K-S statistic is also bounded between this

Name	$\log(P/\text{days})$	e	Original Reference	Name	$\log(P/\text{days})$	e	Original Reference
J0453+1559	0.609808	0.113	Martinez et al. (2015)	J0737-3039	-0.991400	0.088	Kramer et al. (2006)
B1534+12	-0.375718	0.274	Fonseca et al. (2014)	J1756-2251	-0.494850	0.181	Faulkner et al. (2005)
B1913+16	-0.490797	0.617	Hulse & Taylor (1975)	J1913+1102	-0.686133	0.090	Lazarus et al. (2016)
J1757-1854	-0.735182	0.606	Cameron et al. (2018)	J1518+4904	0.936212	0.249	Janssen et al. (2008)
J1811-1736	1.273672	0.828	Corongiu et al. (2007)	J1829+2456	0.070407	0.139	Champion et al. (2004)
J1930-1852	1.653791	0.399	Swiggum et al. (2015)	J1753-2240	1.134751	0.304	Keith et al. (2009)
J1411+2551	0.417638	0.169	Martinez et al. (2017)	J1946+2052	-1.107905	0.064	Stovall et al. (2018)

Table 4.2: The systems we use in our study. These data were originally compiled in [Vigna-Gómez et al. \(2018\)](#).

region. A K-S statistic of 1 means that the compared distribution diverges perfectly from the observed distribution, whereas a K-S statistic of 0 means it matches perfectly.

4.4.1

Pulsar dataset

We use the same dataset of 81 pulsars from [Willcox et al. \(2021\)](#), and filter the dataset in the same way that they do. We remove millisecond pulsars, pulsars in globular clusters, and pulsars known to be in binaries, in order to homogenise the dataset. The PDF and CDF of [Willcox et al. \(2021\)](#), as well as those generated from [Hobbs et al. \(2005\)](#) and [Bray & Eldridge \(2018\)](#), is displayed in Figure 4.3.

The [Hobbs et al. \(2005\)](#) kick has a closed-form expression for the PDF $P(v_k)$, given by Equation 4.3. Consequently, it has a smooth curve in Figure 4.3, whereas the [Bray & Eldridge \(2018\)](#) and [Willcox et al. \(2021\)](#) PDFs are presented as histograms as they are computed.

Figure 4.4 shows the K-S statistic as a function of α and β across our two parameter spaces. We also show the K-S statistic of the Hobbs kick, which has a value of 0.34. We find in general a valley of models with good agreement to [Willcox et al. \(2021\)](#), and find that the global minimum K-S statistic is at $\alpha = 100, \beta = -180$ – agreeing with the prior work of [Bray & Eldridge \(2018\)](#) (who find $\alpha = 100, \beta = -170$).

We also show in Figure 4.5 a selection of four PDFs, alongside their associated CDF and the CDFs for [Willcox et al. \(2021\)](#) and the Hobbs kick. We note that the numerical values shown in Figure 4.5 differs somewhat from those shown in [Richards et al. \(2023\)](#). This was due to a minor bug in the code incorrectly labelling the KS statistics for the figure. The actual result of the paper does not change with the correction of the bug, however the values reported in Figure 5 of [Richards et al. \(2023\)](#) do change.

The four values we present on the left side of Figure 4.5 correspond to $\alpha = 100 \text{ km s}^{-1}, \beta = -170 \text{ km s}^{-1}$ for the fiducial Bray kick, $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ to represent the system with the highest $\ln \mathcal{K}$ value from subsection 4.3.1, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ the lowest, and $\alpha = 40 \text{ km s}^{-1}, \beta = 0 \text{ km s}^{-1}$ the system with low α and zero β , corresponding to a kick that would *a priori* generate logical velocities in the USSNe limit.

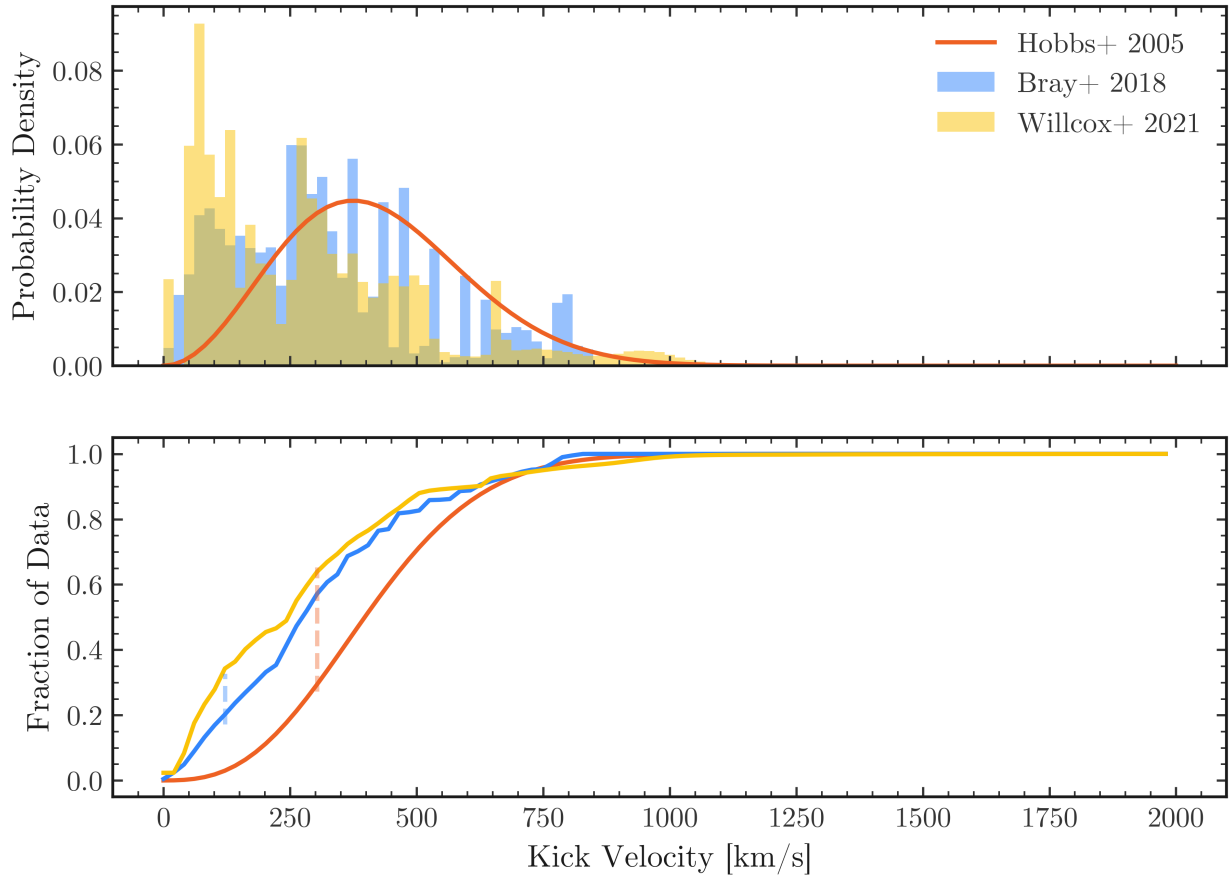


Figure 4.3: The PDF and CDF of the distributions using kicks from [Hobbs et al. \(2005\)](#); [Bray & Eldridge \(2018\)](#); [Willcox et al. \(2021\)](#). [Hobbs et al. \(2005\)](#) has a closed-form expression, whereas the others are computed empirically, hence the Bray and Willcox CDFs are less smooth than that of Hobbs. The two dashed lines indicate where and how the K-S statistic is measured for the associated distribution.

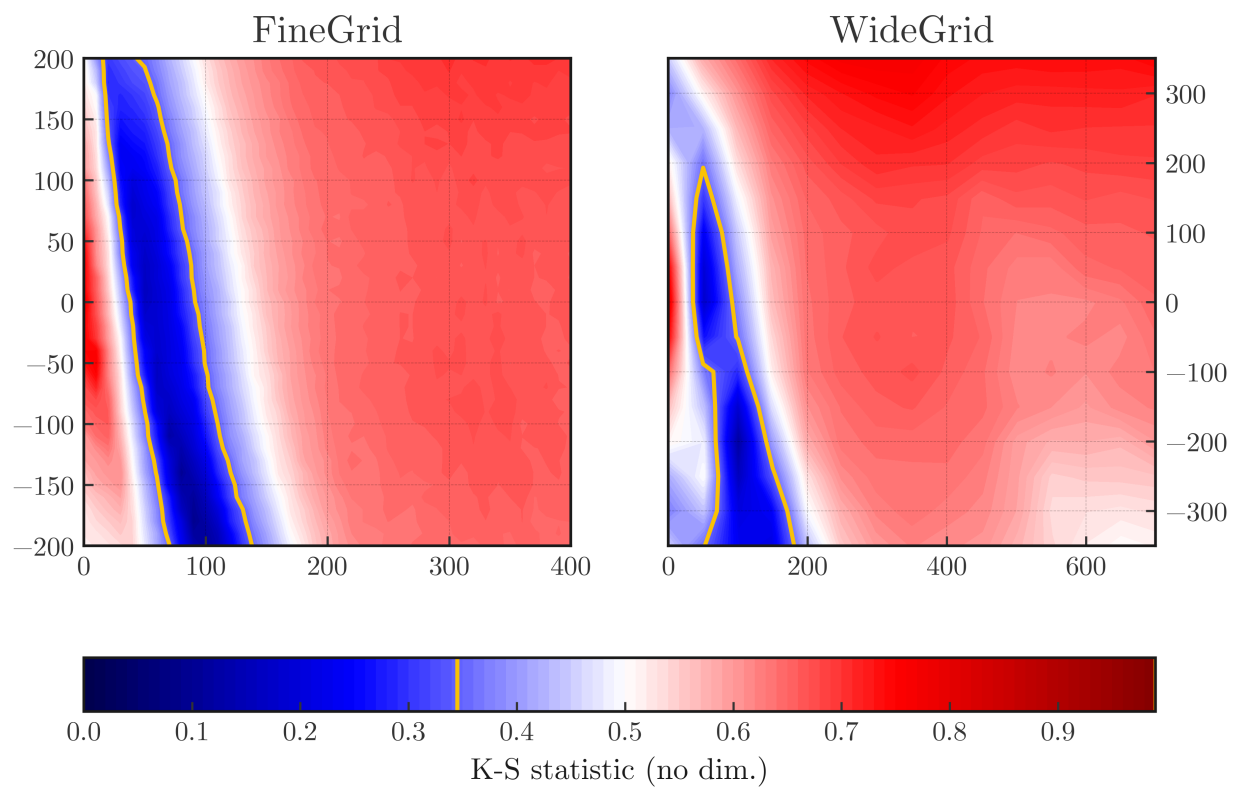


Figure 4.4: The K-S statistic for FINEGRID and WIDEGRID. The orange line denotes the K-S statistic for the Hobbs kick, which has a value of 0.34.

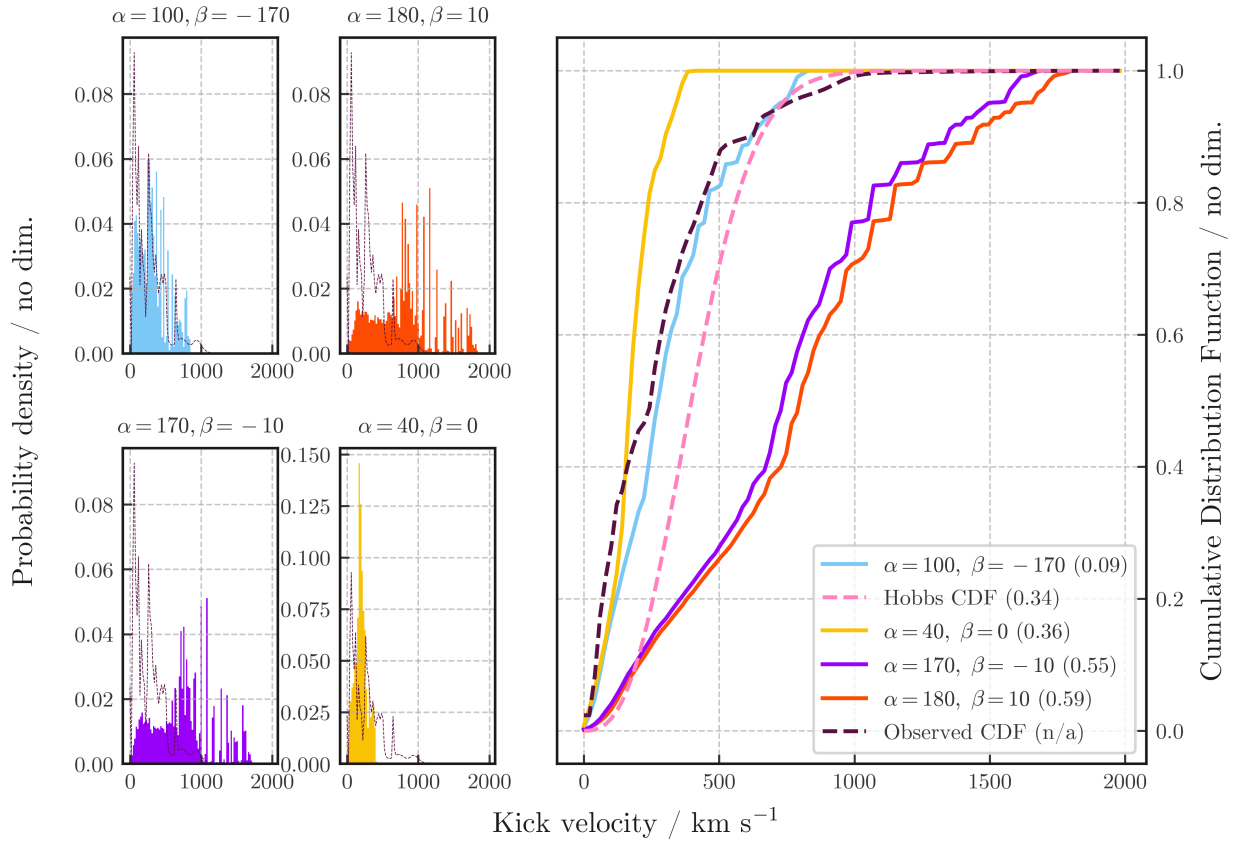


Figure 4.5: The PDFs and CDFs of four systems, alongside the Hobbs CDF and Willcox et al. (2021) CDF. The dashed line on the PDF plots is the PDF of the observed distribution from Willcox et al. (2021). On the right hand plot, the number in parentheses represents the K-S statistic for the distribution, measured relative to Willcox et al. (2021), which by definition has no K-S statistic.

As discussed earlier, the fiducial values of the Bray kick from [Bray & Eldridge \(2018\)](#) admit some problematic extremal cases. If $m_{\text{ej}} \rightarrow 0$, as is the case in a USSN, the kick velocity would become negative. For instance, an “average” USSN with $m_{\text{ej}} = 0.15 M_{\odot}$ and $m_{\text{rem}} = 1.45 M_{\odot}$ would have a kick velocity of $\sim -159 \text{ km s}^{-1}$. The concern here is twofold. Firstly, a negative kick velocity would mean that the velocity vector of the kick would be antialigned with the direction of the more massive remnant; this is a violation of conservation of momentum. Secondly, a kick with high magnitude means that, irrespective of the alignment of the kick, the kick itself is strong. As discussed in [Willcox et al. \(2021\)](#), USSNe are associated with weak kicks, peaking at $v_{\text{k}} \sim 30 \text{ km s}^{-1}$. This means that the fiducial Bray kick is unable to correctly account for USSNe.

To account for these supernova kicks, we introduce a new constraint onto our parameters. Following [Tauris et al. \(2015\)](#), we define a USSN as follows:

1. The ejecta mass is in the range $0 \leq m_{\text{ej}} \leq 0.3 M_{\odot}$, admitting the USSN SN2019dgc from [Yao et al. \(2020\)](#),
2. The remnant mass is in the range $1.1 \leq m_{\text{rem}} \leq 1.8 M_{\odot}$,
3. The envelope mass $m_{\text{env}} \leq 0.2 M_{\odot}$.

For every combination α, β , we record whether it is capable of producing one USSN. The constraint is that this heuristic is met. This constraint is extremely conservative, insofar as we do not constrain on the prevalence of USSNe in our sample, but rather by whether or not they are possible at all. The reason for the conservatism here is that the rates of Type Ic USSNe are expected to be very low. In the BPASS dataset, they comprise 1.8 per cent of all CCSNe ([Stevance, 2022](#)). The paucity of data alongside the paucity of observational evidence suggests a low rate, but since the prevalence is unknown, we place the conservative constraint of at least one USSN to be agnostic. This effectively widens our error bars such that when the USSNe rate is refined further, it will shrink the parameter space.

The constraints above can be overlaid onto our parameter space, which is shown in Figure 4.6. The center of mass of our fit region is located at $\alpha = 117_{-51}^{+36} \text{ km s}^{-1}$, $\beta = 13_{-14}^{+8} \text{ km s}^{-1}$, however this value is computed on a finer grid than the parameter space and is thus likely to be an overfit to the data. We thus report the final recommended model value to the nearest half-bin-width of the parameter space, broadening error bars where necessary, and find $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$ as the recommended model value. Error bars are always broadened here; traditional rounding would result in the upper bound on α (being 153 km s^{-1}) becoming 150 km s^{-1} , which artificially removes potentially permissible regions of the parameter space. Broadening the bars removes this at the cost of potentially admitting values that do not belong within the “acceptable” zone. Further constraints, however, will shrink the region, so this is an acceptable tradeoff.

In Figure 4.7, we present the two major inferred properties of our recommended model value. First, we present the CDFs of a population governed by a Bray kick with the recommended parameters, contrasted

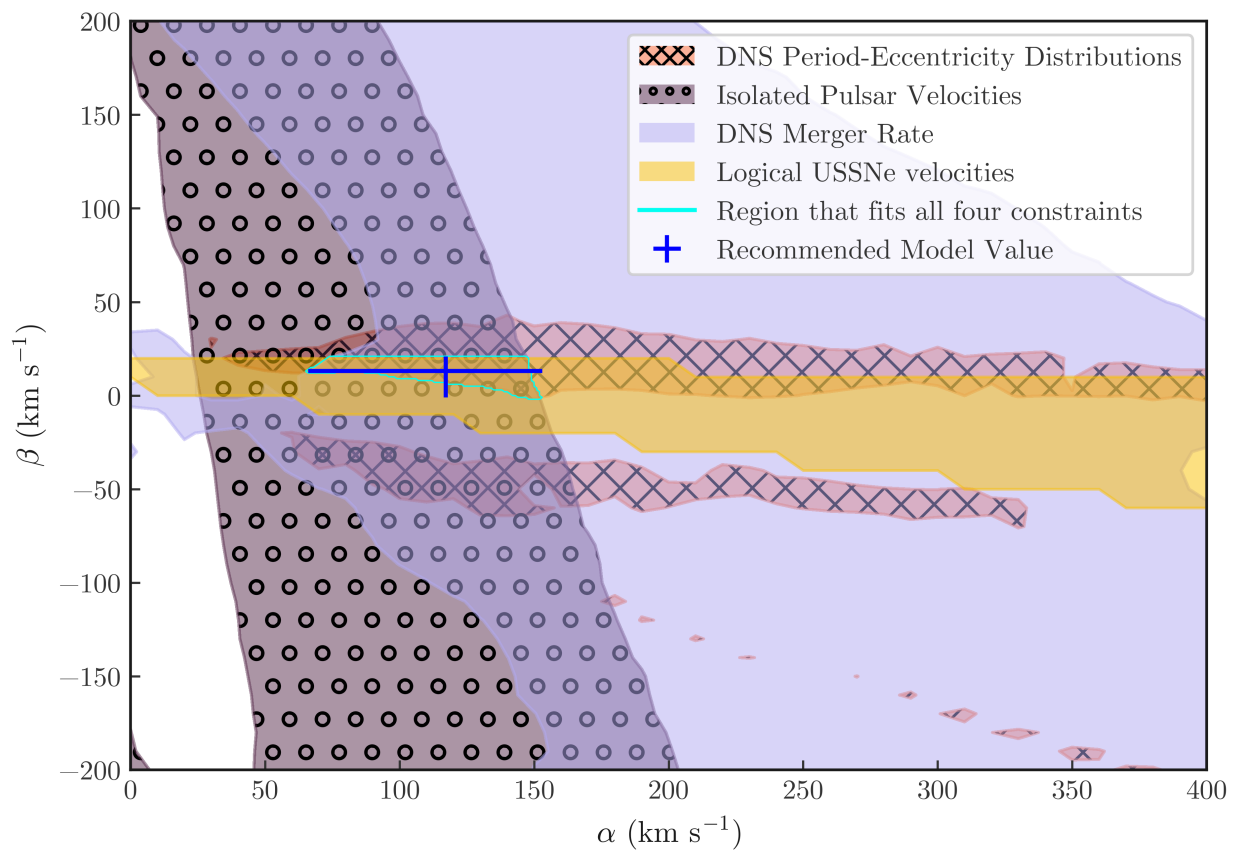


Figure 4.6: The four constraints we apply overlaid on our FINEGRID parameter space. The cyan region represents the region which fits all four parameters, and the blue cross represents the “center of mass” of this region.

with the associated Hobbs kick and observed [Willcox et al. \(2021\)](#) data. We also show the period-eccentricity distribution of the population, with the observed systems from Table 4.2 overlaid atop the distribution.

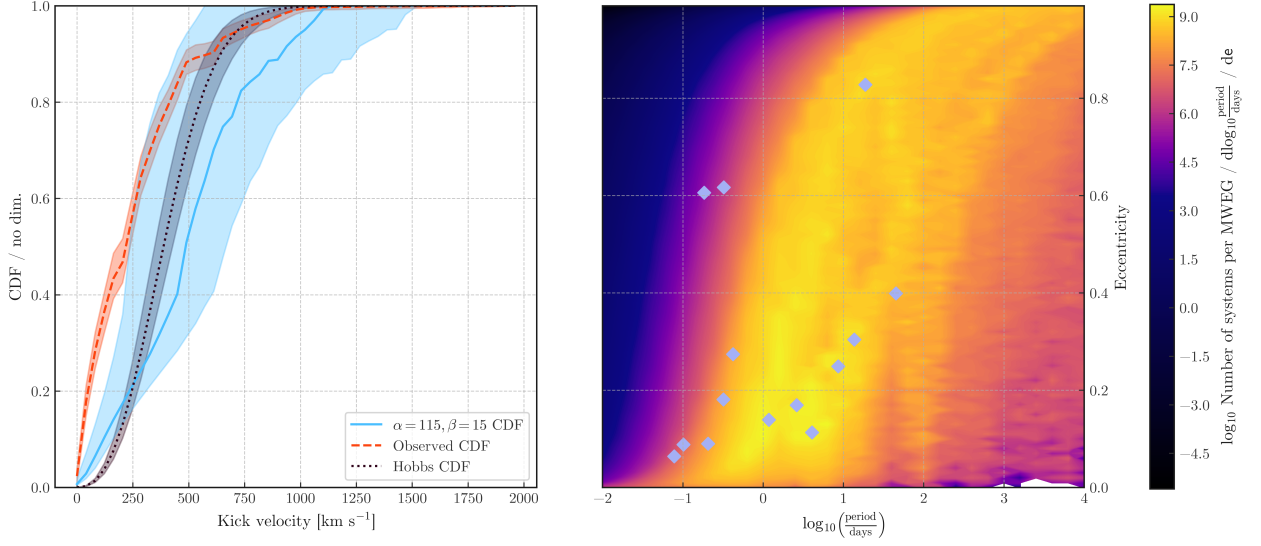


Figure 4.7: The CDF for a population of isolated pulsars (left) and the period-eccentricity distribution (right) for $\alpha = 115 \text{ km s}^{-1}, \beta = 15 \text{ km s}^{-1}$. The blue region on the CDF plot shows the uncertainty region for the reported value. The red region shows the uncertainty band on the [Willcox et al. \(2021\)](#) CDF, which was computed by sampling the PDF 10,000 times, taking 2,000 samples each time, and plotting the extent of the simulated distribution. The black band is the uncertainty on the Hobbs distribution, taking $\sigma_{\text{max}} = 291 \text{ km s}^{-1}$ and $\sigma_{\text{min}} = 239 \text{ km s}^{-1}$ from [Verbunt et al. \(2017\)](#).

We are also able to extrapolate this to comment on the expected peak of the event-rate distribution. Given that the event rate is a convolution between the merger rate of our population and the SFRD we choose (c.f. Equation 2.59), by affixing the SFRD across our populations, by Equation 2.55 we can vary $\mathcal{R}(t)$ by varying α and β . Since the SFRD is of a Gaussian form (with a peak at $z \sim 1.863$), we can also extract the peak of the distribution assuming it is governed by a particular α - β combination. The general results across the parameter space are shown in Figure 4.8. The best model choice parameters we find fall within the valley of $z = 0$, indicating that the peak of the event rate distribution is the present day.

Unfortunately, there is a paucity of observational evidence to confirm whether or not the peak is truly the present day. Given that a redshift of $z = 2$ corresponds to a lookback time of $t_{\text{LB}} = 10 \text{ Gyr}$, we lack evidence to corroborate this prediction. The next generation of telescopes, including JWST, may yet be able to constrain this result further. In the interim, indirect evidence may be able to be applied via the r-process enrichment history of the Milky Way. [Van de Voort et al. \(2022\)](#), for example, finds that greater velocity natal kicks would be correlated with fewer r-process enhanced stars in the Milky Way, as higher kicks would unbind more binaries, meaning that r-process elements are no longer able to accrete onto star-forming regions.

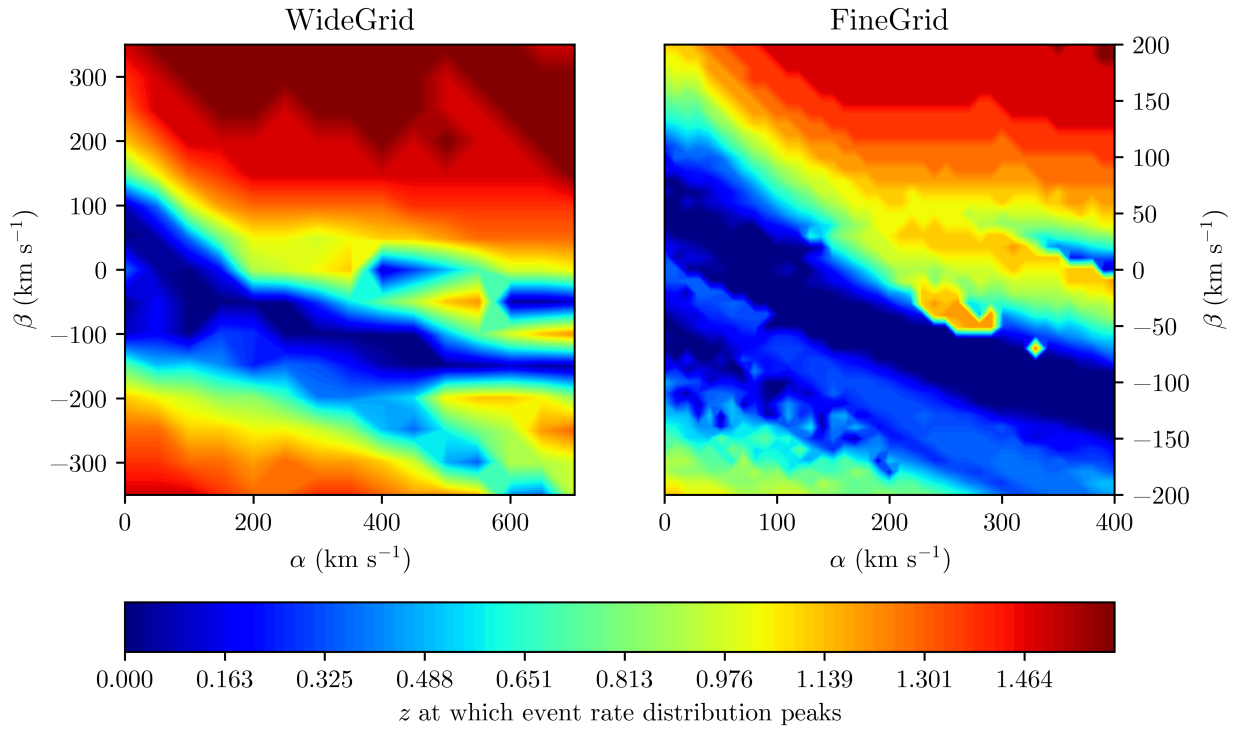


Figure 4.8: The peak of the event rate $\mathcal{R}(t)$ as a function of α and β . Varying these two parameters generally brings the peak closer to the present day ($z = 0$). Absent a merger rate (or, more formally, assuming the merger rate is instantaneous and given by a Dirac-delta function), the SFRD peaks at $z \sim 1.863$.

We have presented a formalism for evaluating and quantifying the goodness-of-fit of the natal kick mechanism against four independent tests. The tests we have presented – constraining the merger rate of BNS systems, the period-eccentricity distribution of galactic BNS systems, the kick velocities of single-star pulsars, and the production of ultra-stripped supernovae – greatly constrain the parameter space to be less than 1 per cent of the possible parameters. Furthermore, the data suggest that the peak of the merger rate distribution is the present day, with $z \simeq 1.5$ as the upper bound of the entire parameter space, and $z = 0$ for our recommended value.

Conclusion

*Oh! I have slipped the surly bonds of Earth
And danced the skies on laughter-silvered wings;*

...

*And, while with silent lifting mind I've trod
The high untrespassed sanctity of space,
Put out my hand, and touched the face of God.*

High Flight, BY JOHN MAGEE JR.

Take-off is optional. Landing is mandatory.

FIRST RULE OF AVIATION

5.1 Summary

5.2 Limitations

*Though in reviewing the incidents of my administration I am
unconscious of intentional error, I am nevertheless too sensible of
my defects not to think it probable that I may have committed many
errors.*

GEORGE WASHINGTON

It is a well-known remark, though attributed to no individual, that Mother Nature is indifferent to humanity. She is also indifferent to our choice of parameters – whilst we can finetune our parameter selection to approximate to essentially arbitrary precision, we have ultimately used **[TODO: ...?]**

5.3 Suggested Future Work

Appendices

Code Samples

Code Sample A.1: The TAKAHE function `integrate_eoms()`

```

1 def integrate_eoms(a0, e0, p):
2     """
3     General purpose integrator for Nyadzani & Razzaque eqns for a & e.
4
5     Params:
6         a0 - The initial semimajor axis, measured in solar radii
7         e0 - The initial eccentricity, dimensionless
8         p - A vector of parameters:
9             p[1] = m1 (units: Solar Mass)
10            p[2] = m2 (units: Solar Mass)
11            p[3] = unused (will be removed in next version)
12            p[4] = unused (will be removed in next version)
13            p[5] = Lifetime (evolution + rejuvenation)
14
15     Returns:
16         A - An array of the semimajor axes of the binary system over time. (Solar
17     ↪ Mass)
18         E - An array of the eccentricities of the binary system over time. (no
19     ↪ dim.)
20     """
21
22     Solar_Mass = 1.989e30          # kg
23     Solar_Radius = 696340000.0    # m
24     G = 6.67e-11                  # m3 kg-1 s-2
25     c = 299792458.0              # m/s
26
27     A = np.array([])
28     E = np.array([])
29     H = np.array([])
30
31     m1, m2, _, _, evotime = p[0], p[1], p[2], p[3], p[4]
32
33     # number of seconds in a year
34     seconds_per_year = 60 * 60 * 24 * 365.25

```

```

33
34 #####
35 #      Unit Check      #
36 #####
37 a = a0 * Solar_Radius # Meters
38 e = e0                 # Dimensionless
39 m1 = m1 * Solar_Mass   # Kilogram
40 m2 = m2 * Solar_Mass   # Kilogram
41 #####
42 #      End Unit Check  #
43 #####
44
45 # Beta has units m4 / s
46 beta = ((64/5) * G**3 * m1 * m2 * (m1 + m2) / (c**5))
47
48 A = np.append(A, a)
49 E = np.append(E, e)
50 H = np.append(H, 0.0)
51
52 total_time = 0
53
54 # Integrate until past the end of the universe, or a 10km orbit
55 while total_time/seconds_per_year + evotime < 1e11 and a > 1e4:
56     # an euler integrator: work out da/dt then times it by dt
57     # to get da, which then we can work out as a = a + da/dt * dt.
58     initial_da = (- beta / ((a**3) * (1 - e**2)**(7/2)))
59     da = initial_da * (1 + (73/24) * e**2 + (37/96) * e**4)
60
61     intial_de = (((-19/12) * beta) / (a**4*(1-e**2)**(5/2)))
62     de = intial_de * (e + (121/304) * e**3)
63     # Units: s-1
64
65     timeA = abs(1e-2 * a/da)
66
67     if e > 1e-10:
68         timeE = abs(1e-2 * e/de)
69     else:
70         de = 0
71         e = 1e-10
72         timeE = timeA * 10

```

```

73
74     # maximum timestep is half of the width of the smallest BPASS time bin
75     conv_frac = 0.23076752*0.5*seconds_per_year
76
77     dt2 = (evotime + total_time/seconds_per_year)*conv_frac
78
79     # Take a timestep that results in the smallest change: either a
80     # change in E, a change in A, 1/2 the smallest BPASS bin.
81     dt = min(timeE, timeA, dt2)
82
83     a = a + dt * da
84     e = e + dt * de
85
86     A = np.append(A, a)
87     E = np.append(E, e)
88     H = np.append(H, dt)
89
90     total_time = total_time + dt
91
92     # Determine why we stopped, so the parent function can tell what went
93     # "wrong".
94     stop_reason = "flag_not_set"
95
96     if total_time/seconds_per_year + evotime >= 1e11:
97         stop_reason = "out_of_time"
98
99     if a <= 1e4:
100         stop_reason = "merged"
101
102     # Solar Radii, Dimensionless
103     return A / Solar_Radius, E, H, stop_reason

```

Code Sample A.2: The TAKAHE function `integrate_timescale()`

```

1 def integrate_timescale(m1, m2, p0, e0):
2     """
3     General purpose integrator for eqn(5.14) of Peters, 1964 [1].
4
5     [1] https://ui.adsabs.harvard.edu/abs/1964PhRv..136.1224P/abstract
6

```

```

7  Params:
8      m1 -- The mass of the first star (Solar Mass)
9      m2 -- The mass of the second star (Solar Mass)
10     p0 -- The initial period (days)
11     e0 -- The initial eccentricity
12
13  Returns:
14     tC -- The coalescence time (seconds)
15  """
16  Solar_Mass = 1.989e30      # kg
17  Solar_Radius = 696340000.0 # m
18  G = 6.67e-11              # m3 kg-1 s-2
19  c = 299792458.0          # m/s
20
21  m1 = float(m1)
22  m2 = float(m2)
23  p0 = float(p0)
24  e0 = float(e0)
25
26  #####
27  #          BEGIN UNIT CHECK          #
28  #####
29  m1 = m1 * Solar_Mass      # Kilograms    ##
30  m2 = m2 * Solar_Mass      # Kilograms    ##
31  p0 = p0 * 24 * 60 * 60    # Seconds      ##
32  e0 = e0                   # Dimensionless ##
33  #####
34  #          END UNIT CHECK          #
35  #####
36
37  # Use Kepler's Third Law to compute the semimajor axis, in meters
38  a0 = (p0**2.0 * (G * (m1+m2))) / (4.0 * np.pi**2.0)**(1.0/3.0) # meters
39
40  # compute the constant c0
41  c0 = a0 * (1.0-e0**2.0) * e0**(-12.0/19.0) * (1.0+(121.0 / 304.0 *
42  ↪ e0**2.0))**(-870.0/2299.0) # meters
43
44  # Compute the constant beta
45  beta = 64.0 / 5.0 * G**3.0 * m1 * m2 * (m1 + m2) / c**5.0
46  # meters4 / s

```

```

46
47     # Circular binary coalescence time
48     tC = a0**4.0 / (4.0*beta)
49
50     if e0 != 0:
51         if e0 < 0.01:
52             # Low ecc - see eqn after eqn(5.14) of Peters, 1964
53             tC = c0**4.0 * e0**(48.0/19.0) / (4.0*beta)
54         elif e0 > 0.99:
55             # High ecc - see eqn after eqn after eqn(5.14) of Peters, 1964.
56             tC = tC * ((768.0 / 425.0) * (1.0-e0**2.0)**3.5)
57         else:
58             # Medium ecc - see eqn(5.14) of Peters, 1964
59             e = 0.0
60             de = e0 / 10000.0
61             summand = 0.0
62
63             while e < e0:
64                 this_integral = de * e**(29.0/19.0)
65                 this_integral *= (1.0 + (121.0/304.0) * e**2.0)**(1181.0/2299.0)
66                 this_integral /= (1.0 - e**2.0)**(1.5)
67
68                 summand += this_integral
69                 e += de
70
71                 e = float(e)
72
73             tC = (12.0/19.0) * (c0**4.0 / beta) * summand
74
75     return tC

```

[TODO: fix label before submission]

Code Sample A.3: The KAITAIKI script required to generate Figure 3.2

```

1 import kaitiaki
2
3 STARS = kaitiaki.STARS.STARSController()
4 STARS.update_run_bs('.')
5
6 generate_ZAMS = False

```

```

7
8 if generate_ZAMS:
9     for target in [8, 10]:
10         STARS.blit()
11         STARS.load_default_modin()
12         STARS.setup_zams_inflation(target)
13
14         STARS.configure_parameters({
15             'DT1': 0.1,
16         })
17
18         STARS.run()
19
20         for f in ['plot', 'out', 'modout']:
21             STARS.terminal_command(f'mv {f} {f}.{target}msun')
22
23 STARS.blit()
24
25 cfgs = ['primary', 'secdry', 'binary']
26
27
28 def run_single(mode):
29     STARS.terminal_command('cp /home/sric560/Desktop/CheckNESI/data data')
30     STARS.setup_single_evolution()
31     STARS.commit_parameters()
32
33     mzams = {'primary': 10, 'secdry': 8}[mode]
34
35     STARS.modout_to_modin(modout_location=f'modout.{mzams}msun',
36                          modin_location='modin')
37
38     STARS.run()
39
40     for f in ['plot', 'out', 'modout']:
41         STARS.terminal_command(f'mv {f} {f}.{mode}')
42
43     print(f"{mode} complete.")
44
45
46 def run_binary():

```

```

47 STARS.terminal_command('cp /home/sric560/Desktop/CheckNESI/data data')
48
49 STARS.setup_binary_evolution()
50 STARS.commit_parameters()
51
52 STARS.modout_to_modin(modout_location='modout.10msun',
53                      modin_location='modin')
54
55 STARS.modout_to_modin(modout_location='modout.8msun',
56                      modin_location='modin2')
57
58 STARS.set_period(30)
59
60 STARS.run()
61
62 for f in ['plot', 'out', 'modout',
63          'plot2', 'out2', 'modout2']:
64     STARS.terminal_command(f'mv {f} {f}.binary')
65
66 print(f"Binary complete.")
67
68
69 for cfg in cfgs:
70     match cfg:
71         case 'primary' | 'secdry':
72             run_single(cfg)
73         case 'binary':
74             run_binary()

```

Period-Eccentricity distributions along the LVK Collaboration O3 contour

In Figure B.1 we display the period-eccentricity distributions along the LVK Collaboration O3 contour. We also display a white dashed line corresponding to a merger time contour of the Hubble time. Most plots have a white region in the lower right corner of the plot, which corresponds to no systems passing through those bins. We use the definition of a Milky-Way Equivalent Galaxy (MWEG) from [Kasen et al. \(2017\)](#), namely that there is one MWEG per $(4.4 \text{ Mpc})^3$.

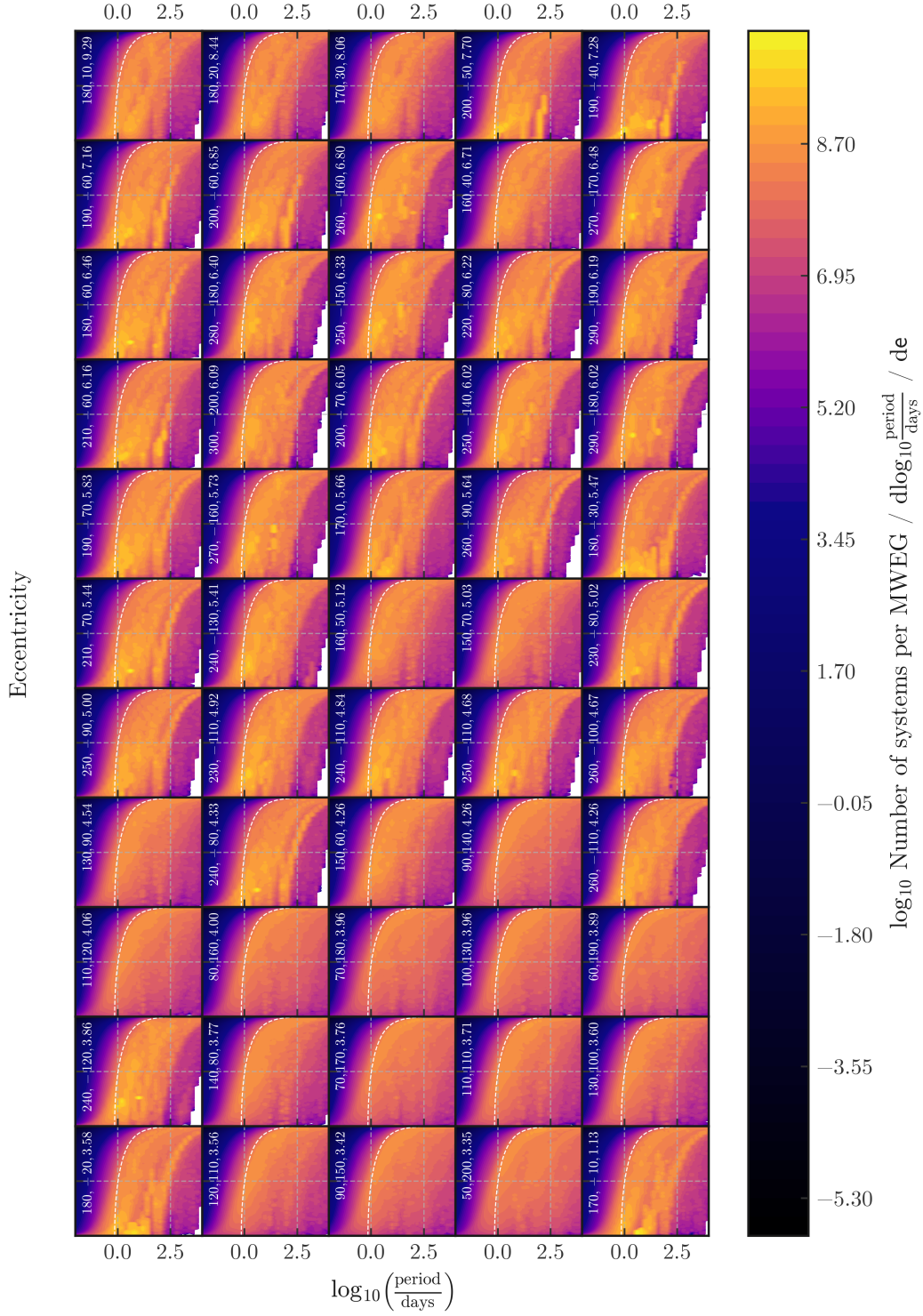


Figure B.1: The period-eccentricity distributions of populations governed by Bray kicks across the LIGO O3 contour. The combination of α and β used in the kick prescription is given in white along the left side of each plot. White regions denote no systems passing through those bins.

Supplementary Maths (if necessary)

Bibliography

- Abbott B. P., et al., 2016, Phys. Rev. Lett., 116, 061102
- Agazie G., et al., 2023, The Astrophysical Journal Letters, 951, L8
- Alighieri D., ca. 1317, Paradiso
- Amaro-Seoane P., et al., 2012, Classical and Quantum Gravity, 29, 124016
- Antoniadis J., et al., 2023, Astronomy & Astrophysics, 678, A50
- Arakaki T., et al., 2023, JuliaPy/pyjulia: v0.6.2, doi:10.5281/zenodo.10443741, <https://doi.org/10.5281/zenodo.10443741>
- Arcavi I., et al., 2017, Nature, 551, 64
- Arnett W. D., Bahcall J. N., Kirshner R. P., Woosley S. E., 1989, ARA&A, 27, 629
- Arzoumanian Z., Chernoff D. F., Cordes J. M., 2002, ApJ, 568, 289
- Asplund M., Amarsi A. M., Grevesse N., 2021, A&A, 653, A141
- Baade W., Zwicky F., 1934, Contributions from the Mount Wilson Observatory, 3, 73
- Baker J. G., Boggs W. D., Centrella J., Kelly B. J., McWilliams S. T., Miller M. C., van Meter J. R., 2008, ApJ, 682, L29
- Ballesteros F. J., 2012, EPL (Europhysics Letters), 97, 34008
- Barstow M. A., Bond H. E., Holberg J. B., Burleigh M. R., Hubeny I., Koester D., 2005, MNRAS, 362, 1134
- Bezanson J., Edelman A., Karpinski S., Shah V. B., 2017, SIAM Review, 59, 65
- Blaauw A., 1961, Bull. Astron. Inst. Netherlands, 15, 265
- Bombaci I., Popov S. B., 2004, A&A, 424, 627
- Bray J. C., Eldridge J. J., 2016, MNRAS, 461, 3747
- Bray J. C., Eldridge J. J., 2018, MNRAS, 480, 5657
- Buck P. H., 1938, Vikings of the sunrise. New Zealand Electronic Text Centre, Wellington, N.Z
- Burbidge E. M., Burbidge G. R., Fowler W. A., Hoyle F., 1957, Reviews of Modern Physics, 29, 547
- Burdge K. B., et al., 2020, ApJ, 905, L7
- Burns J. A., 1978, Vistas in Astronomy, 22, 193

- Böhm-Vitense E., 1958, *Z. Astrophys.*, 46, 108
- Cameron A. D., et al., 2018, *MNRAS*, 475, L57
- Carroll B. W., Ostlie D. A., 2017, *An Introduction to Modern Astrophysics*, 2 edn. Cambridge University Press, doi:10.1017/9781108380980
- Champion D. J., Lorimer D. R., McLaughlin M. A., Cordes J. M., Arzoumanian Z., Weisberg J. M., Taylor J. H., 2004, *MNRAS*, 350, L61
- Christensen N., 1992, *Phys. Rev. D*, 46, 5250
- Christensen N., 2018, *Reports on Progress in Physics*, 82, 016903
- Coppersmith D., Winograd S., 1987, in *Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing. STOC '87*. Association for Computing Machinery, New York, NY, USA, pp 1–6, doi:10.1145/28395.28396, <https://doi.org/10.1145/28395.28396>
- Corongiu A., Kramer M., Stappers B. W., Lyne A. G., Jessner A., Possenti A., D’Amico N., Löhmer O., 2007, *A&A*, 462, 703
- Cox J. P., Giuli R. T., 1968, *Principles of stellar structure*. <https://ui.adsabs.harvard.edu/abs/1968pss.book.....C>
- de Kool M., 1990, *ApJ*, 358, 189
- de Loore C., de Grève J. P., de Cuyper J. P., 1975, *Ap&SS*, 36, 219
- Eddy J. A., Ise R., 1979, *A new sun : the solar results from SKYLAB*. National Aeronautics and Space Administration, <https://ui.adsabs.harvard.edu/abs/1979nssr.book.....E>
- Eggen O. J., 1963, *PASP*, 75, 308
- Eggleton P. P., 1967, *MNRAS*, 135, 243
- Eggleton P. P., 1983, *ApJ*, 268, 368
- Eichler D., Livio M., Piran T., Schramm D. N., 1989, *Nature*, 340, 126
- Einstein A., 1905, *Annalen der Physik*, 322, 891
- Einstein A., 1916, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pp 688–696
- Einstein A., 1917, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pp 142–152
- Eldridge J. J., Stanway E. R., Xiao L., McClelland L. a. S., Taylor G., Ng M., Greis S. M. L., Bray J. C., 2017, *Publ. Astron. Soc. Australia*, 34, e058
- Eldridge J. J., Stanway E. R., Tang P. N., 2019, *MNRAS*, 482, 870
- Fang C., 2015, *Highlights of Astronomy*, 16, 19
- Faulkner A. J., et al., 2005, *ApJ*, 618, L119

Fonseca E., Stairs I. H., Thorsett S. E., 2014, *ApJ*, 787, 82

Fratanduono D. E., et al., 2020, *Phys. Rev. Lett.*, 124, 015701

Friedmann A., 1922, *Z. Phys.*, 10, 377

Fryer C., Burrows A., Benz W., 1998, *ApJ*, 496, 333

Giacobbo N., Mapelli M., 2020, *ApJ*, 891, 141

Gunn J. E., Ostriker J. P., 1970, *ApJ*, 160, 979

Hamacher D., 2022, *The First Astronomers: How Indigenous Elders read the stars*. Allen & Unwin, Crows Nest, New South Wales

Hayashi C., 1961, *PASJ*, 13, 450

Henyey L. G., Lelevier R., Levée R. D., 1955, *PASP*, 67, 154

Henyey L. G., Forbes J. E., Gould N. L., 1964, *ApJ*, 139, 306

Hobbs G., Lorimer D. R., Lyne A. G., Kramer M., 2005, *MNRAS*, 360, 974

Hoyle F., 1946, *MNRAS*, 106, 343

Hoyle F., 1960, *MNRAS*, 120, 22

Hulse R. A., Taylor J. H., 1975, *ApJ*, 195, L51

Hurley J. R., Tout C. A., Pols O. R., 2002, *MNRAS*, 329, 897

Ivanova N., et al., 2013, *A&ARv*, 21, 59

Ivezić u., et al., 2019, *ApJ*, 873, 111

Janssen G. H., Stappers B. W., Kramer M., Nice D. J., Jessner A., Cognard I., Purver M. B., 2008, *A&A*, 490, 753

Jermyn A. S., et al., 2023, *ApJS*, 265, 15

Kasen D., Metzger B., Barnes J., Quataert E., Ramirez-Ruiz E., 2017, *Nature*, 551, 80

Keith M. J., Kramer M., Lyne A. G., Eatough R. P., Stairs I. H., Possenti A., Camilo F., Manchester R. N., 2009, *MNRAS*, 393, 623

Kramer M., et al., 2006, *Science*, 314, 97

LIGO Scientific Collaboration and Virgo Collaboration 2016, *Phys. Rev. Lett.*, 116, 061102

LIGO-Virgo 2021, *ApJ*, 913, L7

LIGO-Virgo-KAGRA 2021, *arXiv e-prints*, p. arXiv:2111.03634

- Lam S. K., Pitrou A., Seibert S., 2015, in Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC. LLVM '15. Association for Computing Machinery, New York, NY, USA, pp 1–6, doi:10.1145/2833157.2833162, <https://dl.acm.org/doi/10.1145/2833157.2833162>
- Langer N., Norman C. A., 2006, *ApJ*, 638, L63
- Lau R. M., et al., 2022, *Nature Astronomy*, 6, 1308
- Lazarus P., et al., 2016, *ApJ*, 831, 150
- Lemaître G., 1931, *MNRAS*, 91, 483
- Levesque E. M., Massey P., Żytkow A. N., Morrell N., 2014, *MNRAS: Letters*, 443, L94
- Madau P., Dickinson M., 2014, *ARA&A*, 52, 415
- Mandel I., 2021, *RNAAS*, 5, 223
- Mandel I., Müller B., 2020, *MNRAS*, 499, 3214
- Martinez J. G., et al., 2015, *ApJ*, 812, 143
- Martinez J. G., et al., 2017, *ApJ*, 851, L29
- Paxton B., Bildsten L., Dotter A., Herwig F., Lesaffre P., Timmes F., 2011, *ApJS*, 192, 3
- Paxton B., et al., 2013, *ApJS*, 208, 4
- Paxton B., et al., 2015, *ApJS*, 220, 15
- Paxton B., et al., 2018, *ApJS*, 234, 34
- Paxton B., et al., 2019, *ApJS*, 243, 10
- Payne C. H., 1925, PhD thesis, Harvard, <https://ui.adsabs.harvard.edu/abs/1925PhDT.....1P>
- Peters P. C., 1964, *Phys. Rev. Lett.*, 136, 1224
- Pian E., et al., 2017, *Nature*, 551, 67
- Poincaré M. H., 1906, *Rendiconti del Circolo Matematico di Palermo (1884-1940)*, 21, 129
- Price D. J., et al., 2018, *Publ. Astron. Soc. Australia*, 35, e031
- Ptolemy C., Proclus ca. 168, *Tetrabiblos*
- Raucq F., Gosset E., Rauw G., Manfroid J., Mahy L., Mennekens N., Vanbeveren D., 2017, *A&A*, 601, A133
- Reardon D. J., et al., 2023, *The Astrophysical Journal Letters*, 951, L6
- Reynolds T. M., Fraser M., Gilmore G., 2015, *MNRAS*, 453, 2885
- Richards S., 2021, Master’s thesis, University of Auckland, doi:10.5281/zenodo.5523933, <https://zenodo.org/record/5523933>

Richards S. M., Eldridge J. J., Briel M. M., Stevance H. F., Willcox R., 2023, MNRAS, 522, 3972

Robertson H. P., 1935, ApJ, 82, 284

Rose W. K., 1998, *Advanced Stellar Astrophysics*. Cambridge University Press

Salpeter E. E., 1955, ApJ, 121, 161

Scherbak P., Fuller J., 2023, MNRAS, 518, 3966

Schmitz K., 2021, JHEP, 2021, 97

Shakespeare W., ca. 1603, *The Tragedy of Othello, the Moor of Venice*

Smartt S. J., et al., 2017, Nature, 551, 75

Stancliffe R. J., Eldridge J. J., 2009, MNRAS, 396, 1699

Stanway E. R., Eldridge J. J., 2018, MNRAS, 479, 75

Stephenson F. R., Yau K. K. C., Hunger H., 1985, Nature, 314, 587

Stevance H., 2022, Private Communication

Stovall K., et al., 2018, ApJ, 854, L22

Sutantyo W., 1978, Ap&SS, 54, 479

Swiggum J. K., et al., 2015, ApJ, 805, 156

Tabarroni G., 1989, MmSAI, 60, 769

Tang P. N., 2024, Private Communication

Tauris T. M., Langer N., Podsiadlowski P., 2015, MNRAS, 451, 2123

Team COMPAS: Riley J., et al., 2022, ApJS, 258, 34

Thorne K. S., Zytlow A. N., 1977, ApJ, 212, 832

ud-Doula A., Sundqvist J. O., Owocki S. P., Petit V., Townsend R. H. D., 2013, MNRAS, 428, 2723

van de Voort F., Pakmor R., Bieri R., Grand R. J. J., 2022, MNRAS, 512, 5258

Various Authors 1611, *The Holy Bible: Conteyning the Old Testament, and the New: Newly Translated out of the Originall Tongues: & with the Former Translations Diligently Compared and Reuised, by His Maiesties Speciall Commandement*. Robert Barker

Verbunt F., Igoshev A., Cator E., 2017, A&A, 608, A57

Vigna-Gómez A., et al., 2018, MNRAS, 481, 4009

Vitense E., 1953, Z. Astrophys., 32, 135

Walker A. G., 1937, Proc. London Math. Soc., 42, 90

Willcox R., Mandel I., Thrane E., Deller A., Stevenson S., Vigna-Gómez A., 2021, ApJ, 920, L37

Wongwathanarat A., Janka H.-T., Müller E., 2013, A&A, 552, A126

Xu H., et al., 2023, Research in Astronomy and Astrophysics, 23, 075024

Yao Y., et al., 2020, ApJ, 900, 46

ISA LEI – NANOQU RARAWA,
Miko sana vodo e na mataka,
Bau nonuma, na nodatau lasa,
Mai Suva, nonuma tikoga.

ISA LEI. THE PURPLE SHADOWS FALL,
Sad the morrow will dawn upon my sorrow.
Ah, forget not when you are far away
Precious moments beside dear Suva Bay.

– *Isa Lei*, “The Fijian Farewell Song”